OMB Number: 4040-0010
Expiration Date: 12/31/2022

APPLICATION FOR FEDERAL ASSISTANCE
SF 424 (R&R)

3. DATE RECEIVED BY STATE	**State Application Identifier**

1. TYPE OF SUBMISSION*

❍ Pre-application ● Application ❍ Changed/Corrected Application

4.a. Federal Identifier
HD059060

b. Agency Routing Number

2. DATE SUBMITTED | **Application Identifier**

c. Previous Grants.gov Tracking Number

5. APPLICANT INFORMATION **Organizational DUNS*:** 8726124450000

Legal Name*: Cornell University
Department: Office of Sponsored Programs
Division:
Street1*: 373 Pine Tree Road
Street2:
City*: Ithaca
County: Tompkins
State*: NY: New York
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 14850-2820

Person to be contacted on matters involving this application
Prefix: First Name*: Mindy Middle Name: Last Name*: McGill-Carlison Suffix:
Position/Title: Grant & Contract Officer
Street1*: 373 Pine Tree Road
Street2:
City*: Ithaca
County: Tompkins
State*: NY: New York
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 14850-2820
Phone Number*: 607-255-2942 Fax Number: 607-255-5058 Email: mmc328@cornell.edu

6. EMPLOYER IDENTIFICATION NUMBER *(EIN) or (TIN)** 1150532082A4

7. TYPE OF APPLICANT* O: Private Institution of Higher Education

Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

8. TYPE OF APPLICATION*

❍ New ❍ Resubmission
● Renewal ❍ Continuation ❍ Revision

If Revision, mark appropriate box(es).
❍ A. Increase Award ❍ B. Decrease Award ❍ C. Increase Duration
❍ D. Decrease Duration ❍ E. Other *(specify)* :

Is this application being submitted to other agencies?* ❍Yes ●No What other Agencies?

9. NAME OF FEDERAL AGENCY*
National Institutes of Health

10. CATALOG OF FEDERAL DOMESTIC ASSISTANCE NUMBER
TITLE:

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT*
REGULATION OF GAMETE USE AND NEURAL PATHWAYS IN REPRODUCTION

12. PROPOSED PROJECT
Start Date* Ending Date*
05/01/2022 04/30/2027

13. CONGRESSIONAL DISTRICTS OF APPLICANT
NY-023

SF 424 (R&R) APPLICATION FOR FEDERAL ASSISTANCE

14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION

Prefix: First Name*: Andrew Middle Name: G Last Name*: Clark Suffix:
Position/Title: Jacob Gould Schurman Professor
Organization Name*: Cornell University
Department: Molecular Biology & Genetics
Division:
Street1*: 227 Biotechnology Bldg.
Street2: 526 Campus Road
City*: Ithaca
County: Tompkins
State*: NY: New York
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 14853-2703
Phone Number*: (607) 255-0527 Fax Number: 607-255-6249 Email*: ac347@cornell.edu

15. ESTIMATED PROJECT FUNDING

a. Total Federal Funds Requested*	$2,001,147.00
b. Total Non-Federal Funds*	$0.00
c. Total Federal & Non-Federal Funds*	$2,001,147.00
d. Estimated Program Income*	$0.00

16.IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?*

a. YES ○ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:
DATE:

b. NO ● PROGRAM IS NOT COVERED BY E.O. 12372; OR
○ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW

17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)

● I agree*

** The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.*

18. SFLLL or OTHER EXPLANATORY DOCUMENTATION File Name:

19. AUTHORIZED REPRESENTATIVE

Prefix: First Name*: Mindy Middle Name: Last Name*: McGill-Carlison Suffix:
Position/Title*: Grant & Contract Officer
Organization Name*: Cornell University
Department: Office of Sponsored Programs
Division:
Street1*: 373 Pine Tree Road
Street2:
City*: Ithaca
County: Tompkins
State*: NY: New York
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 14850-2820
Phone Number*: 607-255-2942 Fax Number: 607-255-5058 Email*: cu_awds@cornell.edu

Signature of Authorized Representative*
Melinda McGill-Carlison

Date Signed*
07/19/2021

20. PRE-APPLICATION File Name:

21. COVER LETTER ATTACHMENT File Name:CoverLetter-ClarkWolfner.pdf

424 R&R and PHS-398 Specific
Table Of Contents

SF 424 R&R Cover Page ... 1
Table of Contents ... 3
Performance Sites ... 4
Research & Related Other Project Information ... 5
Project Summary/Abstract(Description) ... 6
Project Narrative ... 7
Facilities & Other Resources ... 8
Equipment ... 11
Research & Related Senior/Key Person ... 13
PHS398 Cover Page Supplement ... 25
PHS 398 Modular Budget ... 27
Personnel Justification ... 33
PHS 398 Research Plan ... 35
Specific Aims ... 36
Research Strategy ... 37
Progress Report Publication List ... 49
PHS Human Subjects and Clinical Trials Information ... 51
Multiple PD/PI Leadership Plan ... 52
Bibliography & References Cited ... 53
Letters of Support ... 60
Resource Sharing Plan(s) ... 61
Authentication of Key Biological and/or Chemical Resources ... 62

OMB Number: 4040-0010
Expiration Date: 12/31/2022

Project/Performance Site Location(s)

Project/Performance Site Primary Location

❍ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name:	Cornell University
Duns Number:	8726124450000
Street1*:	Molecular Biology & Genetics
Street2:	Rms. 227 & 421 Biotechnology Bldg
City*:	Ithaca
County:	Tompkins
State*:	NY: New York
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	14853-2703
Project/Performance Site Congressional District*:	NY-023

Additional Location(s) File Name:

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED Other Project Information

1. Are Human Subjects Involved?* ❍ **Yes** ● **No**

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ❍ No

If YES, check appropriate exemption number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If NO, is the IRB review Pending? ❍ **Yes** ❍ **No**

IRB Approval Date:

Human Subject Assurance Number

2. Are Vertebrate Animals Used?* ❍ **Yes** ● **No**

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. Is proprietary/privileged information included in the application?* ❍ **Yes** ● **No**

4.a. Does this project have an actual or potential impact - positive or negative - on the environment?* ❍ **Yes** ● **No**

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5. Is the research performance site designated, or eligible to be designated, as a historic place?* ❍ **Yes** ● **No**

5.a. If yes, please explain:

6. Does this project involve activities outside the United States or partnership with international collaborators?* ❍ **Yes** ● **No**

6.a. If yes, identify countries:

6.b. Optional Explanation:

	Filename
7. Project Summary/Abstract*	Project_summary-ClarkWolfner.pdf
8. Project Narrative*	ProjectNarrative-ClarkWolfner.pdf
9. Bibliography & References Cited	BibliographyClarkWolfner.pdf
10.Facilities & Other Resources	Facilities_Resources-ClarkWolfner.pdf
11.Equipment	Equipment-ClarkWolfner.pdf

PROJECT SUMMARY

Sexual reproduction requires precise orchestration of expression of myriad genes in males and females. These genes mediate interactions between the sexes at molecular, cellular, neural and whole-organism levels, including the differential fertility success of sperm from two competing males that mate to a single female. In the current funding period, we used the tractable Drosophila model system to dissect the role of genes expressed in octopaminergic neurons in differential sperm use by a multiply-mated female. In this renewal application, **we propose as Aim 1 a set of experiments that identify genes and quantify their effects on mating plug ejection, a proximal phenotype with a key role in sperm competition outcomes**. Mating plug ejection timing allows females to exert control over paternity of their future offspring. Early ejection of a mating plug disadvantages the male by decreasing the number of his sperm that can be used; conversely longer retention of the plug gives the male a paternity advantage. Both males and females contribute to the mating plug and its ejection. After completing a GWAS for female and male determinants of mating plug ejection timing and validating the resulting genes, we will perform a grid cross to determine whether variation in mating plug timing is mostly determined by genetic variation in the male, or the female, or an interaction between the two. Follow-up perturbation of those genes will begin to unravel genetic pathways linking male quality or its perception with mating plug ejection. **In Aim 2, we will pursue the implications of our recent discovery that mating induces differential exon or promoter use by a large suite of genes in the female brain.** Intriguingly, two of these genes, *desat1* and *foraging,* act in mating discrimination or other reproductive-relevant behaviors. Each strongly activates one of four alternative promoters in response to mating. We will survey the transcriptome to quantify how variation in male factors drive post-mating differential transcript use, how the latter varies across female genotypes, and how this regulation occurs. **In Aim 3 we extend and generalize models designed to quantify and predict outcomes of pairwise interactions.** The classical Bradley-Terry model, widely applied in predicting outcomes of sporting contests between teams that have not yet competed, has a direct analogy in sperm competition “contests.” Assessing fit to models of this class will test whether the males’ fitness can be rank-ordered, and whether those rank orders produce accurate outcome predictions. We will extend these models to the outcomes of the experiments in Aims 1 and 2, and those performed in this project over the years, using the results to assess the overall importance of male x female interactions in each case. Many of the processes that we will study show high levels of evolutionary conservation, implying that our results will expand our understanding of male x female interactions in reproduction that may have relevance to cases of idiopathic human infertility that may involve genetic incompatibility between the partners.

PROJECT NARRATIVE

Fertility requires many levels of interaction between males and females – at the level of gametes and their fates in the reproductive tract and at neural/behavioral levels – and their ultimate success varies dramatically with the genotype of the male and female involved. This variation affects the likelihood of a particular male siring offspring with a given female, due to interactions between male and female components that impact the handling of sperm within the female, and the regulation of genes in her brain. Our proposed studies with a tractable and sensitive model system will uncover the extent and mechanisms by which genetic variation can alter reproductive outcomes, and how these can be predicted analytically for given pairs of interacting animals.

FACILITIES & RESOURCES

Andrew Clark

Laboratory:

The Clark laboratory is located in the Biotechnology Research Building, in the Biology Quad of Cornell's Ithaca campus. The space consists of a suite of six rooms with a total floor area of about 3000 sq ft. The largest laboratory is devoted to molecular biology and contains a laminar flow tissue culture hood. An instrument lab houses two automated 96-well microtiter plate readers, a qPCR instrument, an HP diode-array spectrophotometer, and electrophoretic gel running and documentation facilities. A third lab is dedicated to Drosophila genetics and is set up with 6 dissecting scope stations with CO_2 delivery and generous shelf and table space. In addition, some members of Dr. Clark's computational group are housed in a custom-designed computational suite in 101 Biotech.

The Clark group is nearly equally divided between people who are primarily experimental and people who are primarily computational, and nearly everyone is engaged in both. Dr. Clark's group consists of 2 research associates Elissa Cosgrove and Yassi Hafezi, 3 postdoctoral fellows, Sofie Delbare, Andy Ding (*co-advised with Sumanta Basu*), and Chih-chun Lin; 10 graduate students (Nora Brown (*co-advised with Mariana Wolfner*), Ian Caldas (*co-advised with Philipp Messer*), Dawn Chen (*co-advised with Mariana Wolfner*), Jullien Flynn, Mitch Lokey (*co-advised with Philipp Messer*), Iskander Said (*co-advised with Daniel Barbash*),Tram Nguyen (*co-advised with John Fitzpatrick*); Satyam Srivastav (*co-advised with Cedric Feschotte*); Freddie Xu (*co-advised with Dan Barbash*); You Chen (*co-advised with Haiyuan Yu*); one technician (Asha Jain), and an administrative assistant (Lori Beyea-Powers).

Office:

Dr. Clark's office measures 160 sq ft and is accessed through his laboratory.

Computer:

Each postdoctoral associate and graduate student in our research group has his/her own Core i7- or i9-class or Mac Pro laptop or workstation running a mix of Linux, Windows and OSX. We also have three multi-core servers with disk arrays and a pair of large memory (512 GB) shared-memory multi-core servers. These are connected to the Computational Biology cluster, which consists of an additional 12 48- and 64-core units in a cluster shared with Drs. Charles Danko, Alon Keinan, Philipp Messer, Jason Mezey, Amy Williams, and Haiyuan Yu. We have three disk servers and a total spinning disk capacity of 420 TB. The network connections in the Biotechnology Building operate at 100 Gbps. The Cornell Center for Comparative and Population Genomics (www.3cpg.cornell.edu) hosts short courses on computational analysis of next generation sequencing data for incoming students. Researchers in the Clark's lab also frequently use the computational services offered by the Cornell Bioinformatics Facility (http://www.biotech.cornell.edu/brc/bioinformatics-facility), which provides hardware, software, and consulting support to individual investigators at little or no cost. They quickly install, update and maintain an extensive array of software on the shared BioHPC servers, with more than 250 titles now instantly available.

Mariana Wolfner

Laboratory:

The Wolfner lab is also located in the Biotechnology Building on the Cornell campus. It consists of a suite of 4 rooms. 1200 sq. ft. space is devoted to molecular cell biology and biochemistry and is fully equipped for work with proteins (including extraction and quantification, assays, Western blotting, immunostaining and antibody purification), DNA (including cloning, PCR, CRISPR design, computer analysis, etc.), and RNA (extraction, RT-PCR, etc.). Each of the 8 long-benches in this space ends with a desk for the student/postdoc/staff-member who works at that bench. An Echo/Revolve fluorescence microscope (shared with Clark) is located within this

main lab-space. We also have two rooms (150 sq. ft. each) equipped for fly work, with 3 or 4 microscope-stations, and with incubators for fly-growth at 18°, 25°, and 29°. Across the hall from our main lab is a 1000 sq. ft. heavy equipment room containing equipment (shakers, incubators, etc.) that is shared among several labs. A dishwashing and autoclave room, walk-in 16°, 18°, and 37° incubators, cold rooms, and a shared microscopy room with a GFP-dissection microscope and our Zeiss Axioskop are in a shared microscope-room on our floor, near our lab. The shared staffed Core-facility for Proteomics is in our building.

Dr. Wolfner's current group consists of one postdoctoral fellow (Snigdha Misra), and 7 graduate students (Sarah Allen, Nora Brown (*co-advised with Andy Clark*), Dawn Chen (*co-advised with Andy Clark*), Katie Gordon (*co-advised with Brian Lazzaro*), Jonathon Thomalla (*co-advised with John Schimenti*), Melissa White, and Mengye Yang), one Research Support Specialist (Norene Buehner); 3-5 undergraduate researchers join us during the semesters.

Computers:

Each member of the lab has a Macintosh or PC computer and/or access to our 2 shared iMacs and 2 PCs, along with a scanner and printers. These items are all on a wireless network with cloud-storage and other backups; network connections in the Biotechnology Building operate at 100 Gbps. Additional shared computers, and a poster-printer, are networked in the building. We have all needed software for word and image processing, statistical analysis, DNA and protein sequence analysis, and database searches. Cornell has an extensive software licensing program that maintains a large suite of software including statistical packages such as JMP, and a variety of packages for analysis of DNA sequence, etc. College staff provide IT support.

Office:

Office 160 sq. ft. and is accessed through the lab.

Shared facilities

The Biotechnology Resource Center (BRC) is a staffed core facility in the Institute of Biotechnology (http://www.biotech.cornell.edu/brc/bioinformatics-facility) that provides advanced technology, services, training and educational workshops in Genomics, Imaging, Proteomics, Metabolomics, Flow Cytometry, and Bioinformatics for the Cornell community. These facilities are housed in the first two floors of our building and in the adjacent building, Weill Hall, which can be easily accessed by tunnel. The equipment, reagents, and technical services available through the BRC are frequently utilized by researchers in our laboratory and particularly relevant to Aims 1 and 3 of this proposal. The Biotechnology Building houses several additional professionally-staffed services including an equipment repair facility, Drosophila food kitchen, and dishwashing service.

Shared departmental facilities include autoclave, cell (bacterial and yeast) and tissue (insect and mammalian) culture rooms and facilities, ultracold freezers and high-speed refrigerated centrifuges and ultracentrifuges, and protein purification equipment.

Intellectual environment

We have a wealth of colleagues at Cornell who are able and willing to provide critical advice and materials for our proposed work. The Drosophila research community at Cornell is broad, integrative and collaborative. Within our department there are five other Drosophila labs covering many research specializations, including Daniel Barbash (evolutionary genetics), Charles Aquadro (population genetics), John Lis (biochemistry), Ross MacIntyre (genetics and molecular evolution), Chun Han (CRISPR technologies, and neurodegeneration). Other key Drosophila labs on campus include Brian Lazzaro (immunity) and Nicolas Buchon (stem cells, gut) in Entomology, and David Deitcher and Nilay Yapici (neurobiology and neural development) in Neurobiology and Behavior. All of these labs are in adjacent buildings within the Biology Quad.

The Center for Reproductive Genomics supports state-of-the-art research and connections between basic and translational research in reproductive health. Members of this group of researchers, including John Schimenti, Paula Cohen, Alexander Travis, and Robert Weiss, have expertise in mammalian fertilty and reproduction. The Center supports a seminar series, annual cross-institutional conferences, and training opportunities that are relevant to the reproductive aspects of this proposal.

Finally, computational genomics is particularly strong at Cornell. Both PIs are members of the Cornell Center for Comparative and Population Genomics, comprised of ~25 faculty from across campus (http://3cpg.cornell.edu) that form an energetic and vibrant computational research community. Cedric Feschotte, Amnon Koren, Charles Danko, and Philipp Messer are statistical and computational genomics colleagues that are either located in our building or nearby on campus and have expertise relevant to this proposal.

Animal

N/A (other than Drosophila facilities noted above)

Other

Secretarial and accounting support are provided by the Department of Molecular Biology & Genetics. Staff in our machine shop are available to repair equipment.

EQUIPMENT

Clark Laboratory

Most of the equipment necessary for the proposed project is already in place in the Clark lab or is available as shared departmental or core facilities resources (described below). We have all the standard equipment for DNA, RNA and protein molecular and biochemical analysis, CRISPR genome manipulations, PCR and sequencing, including an ABI Prism 7000 for qRT-PCR, that can be used for verification of extent of knockdowns. We have Drosophila incubators, a Genogrinder for homogenization in 96-well format, various gel electrophoresis equipment, and stereo microscopes for fly sorting. Several NightSea filters and lamps are available for scoring fluorescence in live flies. We have access to Barbash lab's Leica DM5000 microscope with UV epifluoresence, DIC and phase-contrast optics. We also share a hybrid REVOLVE microscope for fluorescent analysis of tissues and live flies with Mariana Wolfner's lab.

Wolfner Laboratory

The lab is well equipped for genetic/molecular/cell biology studies. Examples of equipment available to the lab include: -80° freezer, -20° freezers, refrigerators, incubators for flies (18°, 25°, 29) and bacteria (30°, 37°) microwave oven, gradient PCR, mini-PCR, and other 96-well PCR machines, electrophoresis power supplies and gel boxes, digital gel imaging system, slot-blot apparati, BioRad semi-dry electroblotters, and standard equipment such as waterbaths, balances, pH meters, etc. We have an Echo/REVOLVE microscope (shared with Andy Clark) in the lab; it is equipped with Olympus optics, an iPad/camera, and the capacity for phase and fluorescence microscopy (GFP, fluorescein, rhodamine, DAPI, etc.). We have a Zeiss Axiskop that is also equipped for phase and fluorescence microscopy and with a CCD camera. We have 6 dissecting microscopes for Drosophila work, and share with other Drosophila labs on our floor an additional dissecting scope equipped for GFP visualization and photography. Our lab also has a Leica TCS SP2 confocal, whose 3 lasers (4-line Ar, Green HeNe, and Red HeNe) are fully adjustable by an Acousto-Optic Tunable Filter (AOTF).

Shared Departmental Equipment

Shared departmental large equipment include cell (bacterial and yeast) and tissue (insect and mammalian) culture rooms and facilities, Drosophila food kitchen and microinjection facility, ultracold freezers and high-speed refrigerated centrifuges and ultracentrifuges, and protein purification equipment. A staffed Microscopy and Imaging Facility is available with Zeiss LSM710 and Leica SP2 shared-use laser confocal microscopes in the attached Weill Hall (http://www.biotech.cornell.edu/brc/imaging-facility).

Department of Molecular Biology and Genetics (Drs. Clark and Wolfner's home department): Several shared facilities or equipment including: a staffed dishwashing facility and staffed equipment-repair facility, autoclaves, centrifuges, phosphorimagers and illuminators, LiCOR fluorescence scanner, constant temperature rooms (4°, 16°, 18°, 30°, 37°), and heavy equipment rooms containing all needed incubators and shakers.

Cornell's Proteomics Facility has four advanced LC-MS systems including an UltiMate RSLCnano LC/Orbitrap Eclipse tribrid mass spectrometer, an UltiMateRSLCnano LC/Orbitrap Fusion, an Exion LC/X500B Q-TOF and an UltiMateRSLCnano LC/hybrid triple quadrupole linear ion trap. Both Orbitrap mass spectrometers are used for discovery proteomics analysis in identification and characterization of proteins in relatively complex samples. Both X500B Q-TOF and 4000 QTrap instruments are used for targeted proteomics and targeted metabolomics analyses. The facility has implemented iTRAQ/TMT with the latest Real Time Search SPS-MS3 data acquisition workflow for enhanced quantitative accuracy. The facility has also utilized SILAC, dimethyl labeling techniques, label-free approach and MRM-based targeted analysis for many quantitative proteomic projects, and for characterization of a variety of protein modifications including phosphorylation, glycosylation, acylation, ADP-ribosylation and ubiquitination in complex samples. The Facility has four licensed database searching engines (software) for automated, batch searching of MS and MS/MS data: Proteome Discoverer 2.4 (Thermo Scientific), Mascot 2.5, Byonic 3.5 and MaxQuant 1.6.1

Although our Leica confocal will be perfect for any immunofluorescence needs of this project (we routinely use it to visualize immunofluorescence signals), should we need additional or newer microscopes, the Cornell Imaging Facility has an Olympus epifluorescence scope (DIC optics, high-resolution video and CCD cameras, and MetaMorph Image analysis software) and several newer or more sophisticated microscopes including a

Zeiss LSM710 confocal microscope, two Zeiss LSM880 confocal/multiphoton microscopes, a Zeiss Elyra PS.1 Super Resolution Microscope, an Andor Revolution spinning disk microscope, and a Lavision Biotech Ultramicroscope II light sheet microscope available for use, all on a fee basis.

Cornell's DNA sequencing Core facility is directed by a PhD level scientist (Dr. Peter Schweitzer), and is staffed by talented technicians who help students, postdocs and laboratory staff use core resources. The DNA Sequencing and Genotyping Core houses a NextSeq 2000, 3 NextSeq500's, 3 Illumina MiSeqs, a BioAnalyzer, Life Technologies/ABI QuantStudio3D and Bio-Rad QX100 digital PCR systems, Agilent TapeStation System, a Biosorter, and Biomek and Tecan arraying robots. The RNA Biology Core is run by Dr. Jen Grenier. This facility recently acquired a NanoString Sprint nCounter. Further details about these and other relevant Life Sciences Core Facilities at Cornell can be found at: http://www.biotech.cornell.edu/brc/genomics-facility.

Computer resources

Each postdoctoral associate and graduate student in our research group has his/her own Core i7- or i9-class or Mac Pro laptop or workstation running a mix of Linux, Windows and MacOS. We also have three multi-core servers with disk arrays and a pair of large memory (512 GB) shared-memory multi-core servers. These are connected to the Computational Biology cluster, which consists of an additional 12 48- and 64-core units in a cluster shared with Drs. Charles Danko, Philipp Messer, Jason Mezey, Amy Williams, and Haiyuan Yu. We have three disk servers and a total spinning disk capacity of 420 TB. The network connections in the Biotechnology Building operate at 100 Gbps. The Cornell Center for Comparative and Population Genomics (www.3cpg.cornell.edu) hosts short courses on computational analysis of next generation sequencing data for incoming students. Researchers in the Clark's lab also frequently use the computational services offered by the Cornell Bioinformatics Facility (http://www.biotech.cornell.edu/brc/bioinformatics-facility), which provides hardware, software, and consulting support to individual investigators at little or no cost. They quickly install, update and maintain an extensive array of software on the shared BioHPC servers, with more than 250 titles now instantly available.

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: First Name*: Andrew Middle Name G Last Name*: Clark Suffix:

Position/Title*: Jacob Gould Schurman Professor
Organization Name*: Cornell University
Department: Molecular Biology & Genetics
Division:
Street1*: 227 Biotechnology Bldg.
Street2: 526 Campus Road
City*: Ithaca
County: Tompkins
State*: NY: New York
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 14853-2703

Phone Number*: (607) 255-0527 Fax Number: 607-255-6249

E-Mail*: ac347@cornell.edu

Credential, e.g., agency login: AC347CORNELL

Project Role*: PD/PI Other Project Role Category:

Degree Type: PHD Degree Year: 1980

Attach Biographical Sketch*: File Name: Biosketch-Clark_-_19July2021.pdf

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: First Name*: Mariana Middle Name Federica Last Name*: Wolfner Suffix:

Position/Title*: Professor
Organization Name*: Cornell University
Department: Molecular Biology & Genetics
Division:
Street1*: 423 Biotechnology Bldg.
Street2: 526 Campus Road
City*: Ithaca
County: Tompkins
State*: NY: New York
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 14853-2703

Phone Number*: (607) 254-4801 Fax Number: (607) 255-6249

E-Mail*: mfw5@cornell.edu

Credential, e.g., agency login: mwolfner

Project Role*: PD/PI Other Project Role Category:

Degree Type: PHD,BA Degree Year: 1981,1974

Attach Biographical Sketch*: File Name: NIH_Biosketch_M._Wolfner_6-2021.pdf

Attach Current & Pending Support: File Name:

OMB No. 0925-0001 and 0925-0002 (Rev. 12/2020 Approved Through 02/28/2023)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: **Clark, Andrew G**

eRA COMMONS USER NAME (credential, e.g., agency login): **ac347cornell**

POSITION TITLE: **Jacob Gould Schurman Professor and Nancy and Peter Meinig Family Investigator**

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Brown University, Providence, RI	B.S.	05/1976	Biology/App. Math
Stanford University, Stanford, CA	Ph.D.	08/1980	Population Genetics
Arizona State University, Tempe, AZ	Postdoc	12/1981	Population Genetics
University of Aarhus, Denmark	Postdoc	12/1983	Population Genetics

A. Personal Statement

Dr. Clark has been working on a wide range of problems in population genetics, including many with a direct connection to applications of genomics to human health, since obtaining his PhD at Stanford in 1980. He has published more than 400 peer-reviewed papers in the field of population genetics and is co-author with Dan Hartl of the widely-used textbook *Principles of Population Genetics*. His work is split between efforts in human and Drosophila empirical population genetics, with an emphasis on computationally challenging statistical methods, and on theoretical population genetics, including large simulation studies. He has contributed to the inference of haplotype phase from population genetic samples, to inference of natural selection from genome-wide data, and to inference of past demography from sample genotype data. Many projects have a common theme of relating variation and genotypic and phenotypic levels, which integrates understanding of gene regulatory networks and modern genome-wide approaches to quantitative genetics. Graduate and postdoctoral training has been a major commitment of Dr. Clark, and he has been a successful mentor for 40 PhD students, (10 currently) and 58 postdoctoral fellows, (5 currently), 47 who now hold faculty positions in nine different countries and 22 different states and provinces, and 5 working in industry.

Ongoing and recently completed projects that I would like to highlight include:

R01-GM116113 Wing, Clark, Long (MPI's)
09/08/2016 – 08/31/2021 In NCE
Reference-quality Drosophila Genome Assemblies for Evolutionary Analysis of Previously Inaccessible Genomic Regions

R01 GM119125 Barbash, Clark (MPI's)
03/03/2017 – 02/28/2021 (renewal pending)
Heterochromatin and Satellite Repeat Sequence Variation in Natural Populations

R01 HD059060 Clark, Wolfner (MPI's)
08/24/2009 – 04/30/2022 (renewal pending)
Regulation of Gamete Use and Neural Pathways in Reproduction

R01-GM135926 Basu, Clark (MPI's)
09/23/2019 – 08/31/2023
Learning Dynamics of Biological Processes from Time Course Omics Datasets

R01-ES029929 Ayroles, Zaitlen, Clark (MPI's)
02/01/2019 – 01/31/2024
Improved Methods for Inference of Genotype-Specific Response to Environmental Toxins

R21 AI130635 Messer, Clark (MPI's)
08/21/2017 – 07/31/2020 (project ended)
Improving the Efficiency and Control of CRISPR/Cas9 Gene Drive Systems

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2018-present	Chair, Dept. of Computational Biology, Cornell University, Ithaca, NY
2002-present	Professor, Dept. of Molecular Biology and Genetics, Cornell University, Ithaca, NY
1992-2002	Professor, Dept. of Biology, Pennsylvania State University, University Park, PA
1987-1992	Associate Professor, Dept. of Biology, Pennsylvania State University
1983-1987	Assistant Professor, Dept. of Biology, Pennsylvania State University, University Park, PA

Other Experience and Professional Memberships

2013	Program Committee Chair, American Society of Human Genetics
2012-	Scientific Advisory Board, NHGRI
2012-2015	NIH, Genomes, Comp. Biology and Technology Study Section member
2006-2008	National Human Genetics Research Institute Advisory Council Member
2002-2004, 2006	Genetics Society of America, Council Member, Genetics, Senior Editor
1988-1992, 1994-2002	National Institutes of Health Genetics Study Section (*ad hoc*)

Honors

2018-	Honorary doctorate, Aarhus University, Denmark
2016-	American Academy of Arts and Sciences, elected fellow
2012-	National Academy of Sciences, elected member
2009	Honorary doctorate, University of Oulu, Finland
2006-	Jacob Gould Schurman Professor, Cornell University.
1999-2002	President (elect, acting and past), Society for Molecular Biology and Evolution
1994-present	Fellow, American Association for the Advancement of Science
1992	Alfred P. Sloan Sabbatical Support Award
1987-1992	National Institutes of Health Research Career Development Award

C. Contributions to Science

1. Drosophila heterochromatin and transposable elements

We made our own set of inbred lines of Drosophila of global distribution (Grenier *et al*. 2015) and found an impressive ~800 whole-gene duplications, including several that impact genes in innate immune pathways (Cardoso-Moreira *et al*. 2016). These lines also convinced us that the enormous variation in satellite abundance was a problem worth pursuing seriously, and Wei et al. (2014 *MBE*) first developed the kmer based approach for Illumina read inference of satellite composition. This was extended to multiple species by Wei et al. (2018 *MBE*), showing that satellites turn over with impressive speed across species. Two mutation-accumulation experiments (in Chlamydomonas and Daphnia) confirmed that the rate of change in copy number was as fast or faster than microsatellites (Flynn et al. 2018 GBE; Flynn et al. 2017 *Genetics*). The most impressive case of satellite expansion was *D. virilis*, which has a 7mer that constitutes nearly 40% of the genome (Flynn *et al.* 2020 *MBE*). Variation in genome sequences also occurs in transposable elements, and attributes of their selfish nature is key to understanding their evolution (Flynn *et al.* 2020 PNAS; Ågren and Clark 2018 *PLoS Genet*).

a. Flynn JM, Long M, Wing RA, **Clark AG**. 2020. Evolutionary dynamics of abundant 7 bp satellites in the genome of *Drosophila virilis*. Mol Biol Evol. 37:1362-1375. doi: 10.1093/molbev/msaa010. PMID: 31960929.

b. Flynn JM, Hubley R, Goubert C, Rosen J, **Clark AG**, Feschotte C, Smit AF. 2020. RepeatModeler2 for automated genomic discovery of transposable element families. Proc Natl Acad Sci U S A. 117:9451-9457. doi: 10.1073/pnas.1921046117. PMID: 32300014.
c. Ågren JA, **Clark AG**. 2018. Selfish genetic elements. PLoS Genet 14(11):e1007700. PMID: 30439939; PMCID: PMC6237296.
d. Wei KH, Lower SE, Caldas IV, Sless TJS, Barbash DA, **Clark AG**. 2018. Variable rates of simple satellite gains across the Drosophila phylogeny. Mol Biol Evol. 35:925-941. PMID: 29361128; PMCID: PMC5888958.

2. Drosophila sperm competition

Simple double mating experiments demonstrate that there is enormous variation among lines of *D. melanogaster* for the ability of different males to accomplish sperm competition. We mapped the variation and found that a large portion of the difference maps to genes that encode male seminal proteins (Clark *et al.* 1995 Genetics 139:189-201). We continued to map and characterize both the male and female components of the sperm usage phenotypes (Clark and Begun 1998 Genetics 149:1487-1493), and subsequently teamed up with Dr. Mariana Wolfner to explore the functional aspects of genetic variation in these traits (McGraw *et al.* 2008 Genetics 179:1395-1408). This work continues, with a current emphasis on female receptors of male proteins, and their role in neuronal functioning in the female (Chow *et al.* 2013 *Genetics*; Chen et al. 2019 *Genetics*), and a greatly expanded RNA-seq analysis of post-mating transcriptional changes impacted by both male and female genetic variation (Delbare *et al., 2017*). The *D. novamexicana/D. americana* sister species have nearly zero pre-mating isolation, but strong post-copulatory isolation (and post-copulatory sexual selection), mating this an ideal system to explore transcriptional differences that might drive the isolation (Ahmed-Braimeh *et al*. 2021).

a. Delbare SYN, Ahmed-Braimah YH, Wolfner MF, **Clark AG**.2020. Interactions between the microbiome and mating influence the female's transcriptional profile in *Drosophila melanogaster*. Sci Rep. 10:18168. doi: 10.1038/s41598-020-75156-9. PMID: 33097776.
b. Ahmed-Braimah YH, Wolfner MF, **Clark AG**. 2021. Differences in postmating transcriptional responses between conspecific and heterospecific matings in Drosophila. Mol Biol Evol. 38:986-999. doi: 10.1093/molbev/msaa264. PMID: 33035303; PMCID: PMC7947788.
c. Chen DS, Delbare SYN, White SL, Sitnik J, Chatterjee M, DoBell E, Weiss O, **Clark AG**, Wolfner MF. 2019. Female genetic contributions to sperm competition in *Drosophila melanogaster*. Genetics 212:789-800. PMID: 31101677; PMCID: PMC6614900.
d. Delbare SYN, Chow CY, Wolfner MF, **Clark AG**. 2017. Roles of female and male genotype in post-mating responses in *Drosophila melanogaster*. J Hered. 108:740-753. PMID: 29036644; PMCID: PMC5892397.

3. Drosophila Y chromosome evolution

With Dr. Bernardo Carvalho, my group has discovered and characterized 16 protein-coding genes on the *D. melanogaster* Y chromosome, and the most recent, *FDY*, jumped onto the Y in a block insertion from chromosome 3 just 500 kya (Carvalho *et al.* 2015). In order to pursue the consequences of variation on the Y chromosome, we constructed a set of Y chromosome replacement lines, all retaining the same well-marked autosomal background. Keegan Kelsey discovered exceptionally wide trans-acting effects on position effect variegation mediated by the Y replacement lines (Kelsey and Clark 2017), and an exceptional degree of trans-acting modulation of gene expression, with some 700 genes showing an influence of Y differences (Kelsey *et al.*, in prep.). Most recently Yassi Hafezi performed a beautiful CRISPR knock-out study to dissect the function of Y-linked protein coding genes in Drosophila (Hafezi et al. 2020 *Genetics*).

a. Hafezi Y, Sruba SR, Tarrash SR, Wolfner MF, **Clark AG**. 2020. Dissecting fertility functions of Drosophila Y chromosome genes with CRISPR. Genetics 214:977-990. doi: 10.1534/genetics.120.302672. PMID: 32098759
b. Kelsey KJP, **Clark AG.** 2017. Variation in Position Effect Variegation Within a Natural Population. Genetics 207:1157-1166. doi: 10.1534/genetics.117.300306. PMID: 28931559.
c. Carvalho AB, Vicoso B, Russo CA, Swenor B, **Clark AG**. 2015. Birth of a new gene on the Y chromosome of *Drosophila melanogaster*. Proc. Natl. Acad. Sci. USA 112:12450-12455. PMID: 26385968.

4. Human population genetics

Population genetics has seen an amazing expansion in the last 30 years, from its origins as a mostly theoretical field, to its vital role in making sense of variation in genome sequence data, especially in human genetics. My first contribution in this area was a paper on the inference of haplotype phase based only on population sample data. This paper (Clark 1990 *MBE* 7:111-122) was the first to recognize that latent in population sample data was information about high-order linkage disequilibrium, and it explored both initial algorithms for inference of phase, but also the constraints on accurate phase inference. Next w examined sequence variation in the LPL locus in 71 people (Clark *et al.* 1998 *AJHG* 63:595-612). The complexity of these data (88 SNPs with a rich haplotype structure), made quite an impression on the field, and this paper established the need to understand fine-scale LD in human genetics. At about the same time I became a consultant with Celera Genomics, and worked on the means by which population genetic principles might impact genome sequence assembly and how we might perform inference of SNPs jointly with the assembly (Venter *et al.* 2001 *Science* 291:1304-1351). With the publication of the mouse genome sequence, the idea of doing genome-wide evolutionary analysis became accessible. Along with Rasmus Nielsen and Carlos Bustamante, we were the first to perform a genome-wide scan for natural selection in the human-chimp-mouse trip (Clark *et al.* 2003 *Science* 302:1960-1963). Subsequently we published several more papers on genome-wide scans for selection. Mostly recently we have employed population and quantitative genetics approaches in a successful collaboration with Dr. Ruth Ley to identify genetic variation in humans that modulates composition of the gut microbiome (Blekhman *et al*. 2015; Goodrich *et al*. 2014, 2016, 2017). A recent collaboration with Haiyuan Yu's lab is dissecting the roles of disrupted protein-protein interaction on gene function and population genetic attributes across large numbers of human single nonsynonymous site variants (Fragoza et al. 2019).

a. Marderstein AR, Davenport ER, Kulm S, Van Hout CV, Elemento O, **Clark AG**. 2021. Leveraging phenotypic variability to identify genetic interactions in human phenotypes. Am J Hum Genet. 108:49-67. doi: 10.1016/j.ajhg.2020.11.016. PMID: 33326753.
b. Marderstein AR, Uppal M, Verma A, Bhinder B, Tayyebi Z, Mezey J, **Clark AG**, Elemento O. 2020. Demographic and genetic factors influence the abundance of infiltrating immune cells in human tissues. Nat Commun. 11(1):2213. doi: 10.1038/s41467-020-16097-9. PMID: 32371927.
c. Fragoza R, Das J, Wierbowski SD, Liang J, Tran TN, Liang S, Beltran JF, Rivera-Erick CA, Ye K, Wang TY, Yao L, Mort M, Stenson PD, Cooper DN, Wei X, Keinan A, Schimenti JC, **Clark AG,** Yu H. 2019. Extensive disruption of protein interactions by genetic variants across the allele frequency spectrum in human populations. Nat Commun. 10:4141. PMID: 31515488; PMCID: PMC6742646.
d. Goodrich JK, Davenport ER, **Clark AG**, Ley RE. 2017. The relationship between the human genome and microbiome comes into view. Annu Rev Genet. 51:413-433. PMID: 28934590; PMCID: PMC5744868.

5. Comparative genomics and transcriptomics

Having whole-genome sequences and whole-transcriptome (RNA-seq) sequences from multiple species affords many opportunities for analysis of evolutionary pressures at play on genomes. We spear-headed a large team in the analysis of the 12 Drosophila species' genome sequences, and laid out a series of questions and methods of analysis in an impactful paper that has since been emulated by several other groups (Clark *et al*. 2007). Quantification of allele-specific expression became a keen interest early on, and we were the first to lay out a clear approach for inference of *cis*- and *trans*-regulation of expression in interspecific hybrids (Wittkopp *et al.* 2004 MBE). Realizing that F1 hybrids among strains within a species can give considerable insight about allelic expression, we turned this design to reciprocal crosses for inference of genomic imprinting (Wang *et al.* 2011 *Genetics* 189:109-122), applying this method first to mice and subsequently branching out to six additional organisms. We showed that the South American opossum, while predominantly inactivating the paternal X, nevertheless also expresses a number of X-linked genes that escape X-inactivation altogether, much like escapers of random X-inactivation, although these two groups of genes are non-overlapping (Wang *et al.* 2014). We also applied allele-specific gene expression and other genomic analysis to the classic tame fox study (Wang et al. 2019 *PNAS*), and to assess the role of genome methylation on gene expression in the parasitoid Nasonia wasps (Wang et al. 2016 *PLoS Biol*).

a. Arguello JR, Laurent S, **Clark AG**. 2019. Demographic history of the human commensal *Drosophila melanogaster*. Genome Biol Evol. 11:844-854. PMID: 30715331; PMCID: PMC6430986.
b. Wang X, Pipes L, Trut LN, Herbeck Y, Vladimirova AV, Gulevich RG, Kharlamova AV, Johnson JL, Acland GM, Kukekova AV, **Clark AG**. 2018. Genomic responses to selection for tame/aggressive behaviors in the silver fox (*Vulpes vulpes*). Proc Natl Acad Sci U S A. 115:10398-10403. PMID: 30228118; PMCID: PMC6187196.
c. Wang X, Werren JH, **Clark AG**. 2016. Allele-specific transcriptome and methylome analysis reveals stable inheritance and cis-regulation of DNA methylation in Nasonia. PLoS Biol. 14(7):e1002500 PMID: 27380029.
d. Chow CY, Wang X, Riccardi D, Wolfner MF, **Clark AG**. 2015. The genetic architecture of the genome-wide transcriptional response to ER stress in the mouse. PLoS Genet. 11(2):e1004924. PMID: 25651210.

Complete List of Published Work in MyBibliography:
https://www.ncbi.nlm.nih.gov/pubmed/?term=andrew+g+clark
https://scholar.google.com/citations?user=qitILSIAAAAJ&hl=en&oi=ao

OMB No. 0925-0001 and 0925-0002 **(Rev. 12/2020 Approved Through 02/28/2023)**

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: **Wolfner, Mariana F.**

eRA COMMONS USER NAME (credential, e.g., agency login): **mwolfner**

POSITION TITLE: **Distinguished Professor of Molecular Biology and Genetics**

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Cornell University, Ithaca, NY	B.A.	05/1974	Biology and Chemistry
Stanford University, Stanford, CA	Ph.D.	01/1981	Biochemistry
University of California, San Diego, La Jolla, CA	Postdoc	07/1983	Molecular Genetics

A. Personal Statement

My lab's research reflects my long-term interest in the molecular mechanisms of how organisms develop and reproduce and how these mechanisms evolve; we generally using *Drosophila* as a model to dissect these questions. As a graduate student, I was central to a group within the Hogness lab that developed and used differential cDNA hybridization approaches to identify steroid-regulated genes in flies. As a postdoc, I studied the basis for sex differences, cloning the *doublesex* (fly DMRT ortholog) gene with my PI Baker. My own lab is fascinated with the reproductive and developmental events that surround the time of fertilization. We combine mutational, RNAi, 'omic, ectopic expression, biochemical, evolutionary, and physiological analyses to dissect conserved reproductive processes such as seminal protein action and egg activation, and the evolutionary phenomena that impact these events. We are fascinated by the idea that molecules from one animal can act in another animal, and about the genetic underpinnings of such phenomena and their variation within species. We have addressed this through studies of male-by-female genotype interactions in sperm preference/competition (in collaboration with A. Clark), and by dissecting the mechanisms by which seminal proteins promote fertility by fostering sperm storage and maintenance, egg production and ovulation, and physiological and neural/behavioral changes that promote fertility and sperm preference. Our results have taken us in additional directions, including studies of the development of male reproductive glands and their cell types, and (with L. Harrington) the actions of seminal proteins in insect vectors of disease. Our other research area concerns the conserved events of the egg-to-embryo transition. We identified and studied genes whose conserved products help "activate" the fertilized egg to foster embryo development. We study the triggers of this process, discovering that *Drosophila* egg activation is triggered by mechanical forces that open ion channels to let calcium into the oocyte which, in turn, catalyzes changes to the oocyte proteome to start development.

Our research stems from the work of the superb 43 graduate students, 40 postdocs, 91 undergrads from diverse backgrounds, and 1 staff member whom I've been fortunate to recruit to my lab, and whose mentoring and training I consider extremely important and rewarding. Most of my graduate and postdoc mentees have taken academic positions; others hold positions at the NIH, in industry, or in science publishing/communication.

Ongoing projects that I would like to highlight include:

R01 HD059060
Clark (PI), Wolfner (PI)
08/01/2017 – 04/30/2022

Regulation of gamete use and neural pathways in reproduction

R37 HD038921
Wolfner (PI)
09/15/20-06/30/25
Actions of seminal proteins in mated Drosophila females

R01 AI095491
Harrington (PI), Wolfner (PI)
02/01/2017 – 01/31/2022
New targets for reproductive control of mosquito vectors

R03 HD101732
Wolfner (PI)
09/26/20-08/31/22
Determining the role of the conserved TRPM ion channel in egg activation, using the Drosophila model

R21 HD105230
Wolfner (PI), Schimenti (PI)
04/01/21 - 03/31/23
Genetics and proteomics of mouse egg activation

Citations (from total of 217; * = co-corresponding authors):

Chapman, T., Liddle, L., Kalb, J.M., Wolfner, M.F. and Partridge, L. (1995). Cost of mating in *Drosophila melanogaster* female is mediated by male accessory gland products. Nature 373: 241-244. PMID: 7816137.

Heifetz Y. and Wolfner M.F. (2004). Mating, seminal fluid components, and sperm cause changes in vesicle release in the *Drosophila* female reproductive tract. Proc. Natl. Acad. Sci. USA 101, 6261-6266. PMC395957

Misra, S. and Wolfner, M.F. (2020) *Drosophila* seminal Sex Peptide associates with rival as well as own sperm, providing SP function in polyandrous females. eLife, 9:e58322. PMC7398695

Hu, Q., Duncan, F.E., Nowakowski, A.B., Antipova, O.A., Woodruff, T.K., O'Halloran, T.V. and Wolfner, M.F. (2020) Zinc dynamics during Drosophila oocyte maturation and egg activation. iScience, 23(7):101275. PMC7330606

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

1995-present	Professor (and Assoc. Chair 2001-2005, 2021-pres.); Dept. of Molecular Biology and Genetics, Cornell
1991-92, 1994-96	Director of Grad. Study, Field of Genetics and Development, Cornell Univ., Ithaca, NY
1989-1995	Associate Professor; Genetics and Development, Cornell University, Ithaca NY
1983-1989	Assistant Professor; Genetics and Development, Cornell University, Ithaca NY

Other Experience

2019-'21, 1990-'92	National Drosophila Board (President, 1994; 2021); committee member or chair
2015, 2017	Vice Chair (2015), Chair (2017) Gordon Conf. on Fertiliz. and Activ'n of Development
2007-2009	Chair (elect/active/retiring) Biological Sciences Section G, AAAS
2006-2013	Member, Board of Directors ('08-'08), Secretary ('10-'13) Genetics Soc. of America
2001	42nd National Drosophila Research Conference: co-Chair of organizing committee
2001-present	Editorial Boards (-pres. unless noted): PLoS Biology, PNAS, Insect Biochem. Mol. Biol., GENETICS, FLY, J. Ins. Sci. ('03-'19), Spermatogen. ('10-'17), Mol. Repro. Devt. ('11-'18)
1993-present	Service on NIH Study Sections or NSF Grants Panels; recent ones include: NIH CMIR (2008-2014; Vice-Chair 2010, 2013); F-05D/U (2016-2018); ZRG1-BMCB-A (2018); ZCA1-SRA-O (2018); VB (2019), ZHD1 DSR-Z (2021)

Honors

2020	MERIT Award, NICHD
2019	Elected to National Academy of Sciences
2019	Certificate of Distinction, from Council of International Congress of Entomology
2018	Genetics Society of America Medal
2017	Recognition Award in Insect Physiology, Biochem. and Toxicol., Entomol. Soc. of America
2003-present	Distinguished (formerly Goldwin Smith) Professor of Molecular Biology and Genetics, Cornell University
2012	Kendall S. Carpenter Memorial Award for Distinguished Advising, Cornell University
2010	Lady Davis Fellow, Hebrew Univ. of Jerusalem
2009	Steven and Margery Russell Award for Distinguished Teaching, Cornell University
2008-present	Distinguished lectures including: Huck Institutes, Penn State U.; Wilhelmine Key Lecturer, Amer. Genetics Assn.; Nelson Lecture, U. of Missouri; Al Downe Lecture, Queen's U.; Keynote at Annual Drosophila Conference; Distinguished Women in Science Lecture, Monash University, Australia; Kathleen Osborn Lecture, University of Kansas Medical Center; Christiana Smith Lecture, Mount Holyoke College; M.C. Chang Lecture, University of Massachusetts Medical School
2006-present	AAAS Fellow
2006	Robert A. and Donna B. Paul Award for Advising Excellence, Cornell University
2003-present	Stephen H. Weiss Presidential Fellow, Cornell University
1983-1998	DuPont Young Faculty Award; Cornell Biotechnology Institute Young Investigator Grant/Award; Basil O'Connor Starter Research Scholar, March of Dimes; Career Advancement Award, National Science Foundation; Faculty Research Award, American Cancer Society; POWRE Award, NSF

C. Contributions to Science

1. Insect seminal proteins are important in reproduction

Seminal proteins are now recognized as important for reproductive processes like sperm storage and utilization (insects, mammals), uterine contraction (insects, mammals), progeny quality (mice), ovulation (insects, camelids), behavioral changes (insect feeding and mating behaviors). These hormones or inducers made in the male influence the female's physiology. We identified the full suite of *Drosophila* seminal proteins, and used genetics to identify those that cause specific post-mating effects in female flies. These include an ovulation-inducer, a molecule that modulates uterine contraction, one that affects gut physiology, and several that affect sperm storage/retention in females. We identified biochemical pathways in *Drosophila* seminal fluid analogous to ones in mammals, such as a proteolytic pathway that processes seminal proteins, and one that binds a seminal protein to sperm. We identified cell types and functions of the tissue that makes seminal proteins. We extended our work to an insect of medical importance, by examining the nature and functions of seminal proteins in *Aedes* mosquitoes that vector arboviruses such as dengue, zika, and Chikungunya viruses. Beyond our research findings, I have played major roles in the community, organizing biennial international Insect Reproductive Molecules meetings, as well as conferences or conference-sessions in related areas.

Kalb, J., DiBenedetto, A.J. and Wolfner, M.F. (1993) Probing accessory gland function by directed cell ablation in *Drosophila*. Proc. Natl. Acad. Sci. USA 90: 8093-8097. PMC47294

Mueller, J.L., Ripoll, D.R., Aquadro, C.F. and Wolfner, M.F. (2004) Comparative structural modeling and inference of conserved protein classes in Drosophila seminal fluid. Proc. Natl. Acad. Sci. USA, 101,13542-7. PMC518759

Degner, E.C., Ahmed-Braimah, Y.H., Borziak, K., Wolfner, M.F.*, Harrington, L.C.*, and Dorus, S.* (2019) Reproductive functions and genetic architecture of the seminal fluid and sperm proteomes of the mosquito *Aedes aegypti*. Mol. Cell. Proteomics 18, S6-S22. PMC6427228

White, M.A., Bonfini, A., Wolfner, M.F.* and Buchon, N.* (2021) *Drosophila* melanogaster sex peptide is a key regulator of female midgut morphology and physiology. Proc. Natl. Acad. Sci. USA, 118(1):e2018112118. PMC7817148

2. How seminal proteins regulate female reproductive physiology

We identified physiological and molecular mechanisms by which seminal proteins alter female reproductive physiology in *Drosophila* and *Aedes*, including by developing or adapting new techniques, such as micro-CT

scanning and methods to quantify neuromodulators and their action. We discovered that the fly and mosquito female's transcriptome is altered by mating and seminal proteins (as also occurs in mammals). We showed that mated females produce combinations of neuromodulators that coordinate events along the reproductive tract and are, in turn, modulated by mating and seminal proteins. Using micro-CT scanning, we obtained a detailed view of the post-mating changes in the female's reproductive tract. We found new phenomena such as coordinated looping and unlooping of the oviduct that regulates the rate of egg movement through the tract, and that a seminal protein "jumpstarts" this process by relaxing the oviduct's musculature, through regulating the female's octopamine (invertebrate norepinephrine) neuromodulator. We identified the role of the mating plug in *Drosophila* reproduction showing some functions analogous to those of copulatory plugs in mammals, and additional functions in regulating the uncoupling of the mating pair. We discovered behaviors that are regulated by seminal proteins, such as the incidence of harmonic convergence in *Aedes* mosquitoes.

Rubinstein, C.D. and Wolfner, M.F. (2013) The *Drosophila* seminal protein ovulin mediates ovulation through female octopamine neuronal signaling. Proc. Natl. Acad. Sci. USA, 110,17420-5. PMC3311333

Heifetz, Y., Lindner, M., Garini, Y. and Wolfner, M.F. (2014) Mating regulates neuromodulator ensembles at nerve termini innervating the *Drosophila* reproductive tract. Current Biology, 24,731-7. PMC4020355

Mattei, A.L., Riccio, M.L., Avila, F.W. and Wolfner, M.F. (2015) An integrated 3D view of post-mating responses by the *Drosophila melanogaster* female reproductive tract, obtained by micro-CT scanning. Proc. Natl. Acad. Sci USA, 112, 8475-80. PMC4500220

League, G.P., Baxter, L.L., Wolfner, M.F.* and Harrington, L.C.* (2019) Male accessory gland molecules inhibit harmonic convergence during courtship flight in the mosquito *Aedes aegypti*. Current Biology 29, R196-R197. PMC6800996

3. Genetic/genomic interactions between males and females during mating and sperm-competition.

In collaboration with A. Clark, we capitalized on intraspecific sequence variation to identify important molecules in the male x female interactions that underlie sperm competition/preference. The female genes point to a new, active, role for the female's nervous system in sperm competition/preference. We used RNAi and targeted neural manipulation to confirm this, and to identify the genes and neurons involved. Our GWAS/RNAi method generalized to other processes, and we exploited it to identify genes likely important in human disease etiology such as ER stress and retinal degeneration. Finally, we also examined the contribution of genetic variation to the transcriptional changes in mated females, that underlie sperm competition and post-mating responses.

Chow, C.Y., Wolfner, M.F.* and Clark, A.G.* (2013) Large neurological component to genetic differences underlying biased sperm use in Drosophila. Genetics 193,177-85. PMC3527244

Chow, C.Y, Wolfner, M.F.* and Clark, A.G.* (2013) Using natural variation in *Drosophila* to discover previously unknown endoplasmic reticulum stress genes. Proc. Natl. Acad. Sci. USA 110: 9013-8. PMC3670321

Delbare, S.Y.N., Chow, C.Y., Wolfner, M.F.* and Clark, A.G.* (2017) Roles of female and male genotype in post-mating responses in *Drosophila melanogaster*. J. Hered., 8, 740-753. PMC5892397.

Chen, D.S., Delbare, S.Y.N., White, S.L., Sitnik, J.L., Chatterjee, M., Dobell, E.L., Weiss, O.D., Clark, A.G.*, and Wolfner, M.F.* (2019) Female genetic contributions to sperm competition in *Drosophila melanogaster*. Genetics 212, 789-800. PMC6614900.

4. Evolutionary pressures that shape reproductive protein function.

Despite conservation of seminal proteins' actions and biochemical types across species, their primary amino acid sequences are often evolutionarily labile. We showed this in *Drosophila*, but it is a general phenomenon, including in humans. We also discovered that the expression levels of seminal proteins can vary within and between populations, and that new seminal protein genes can arise by duplication and divergence of other reproductive genes. All of this lability is important in reproductive isolation, making seminal proteins a major molecular model in the field of sexual selection. We found that male flies can tailor their ejaculates, donating different amounts or blends of seminal proteins to females depending on presence of competitors. We exploited the evolutionary lability of reproductive proteins to develop "evolutionary EST screening" to identify evolutionarily labile proteins, and evolutionary rate co-variation as a way to screen for new members of protein pathways.

Swanson, W.J., Clark, A.G., Waldrip-Dail, H. Wolfner, M.F. and Aquadro, C.F. (2001) Evolutionary EST analysis identifies rapidly evolving male reproductive proteins in Drosophila. Proc. Natl. Acad. Sci. USA 98, 7375-7379. PMC34676

Sirot, L.K., Wolfner, M.F. and Wigby, S. (2011) Protein-specific manipulation of ejaculate composition in response to female mating status in *Drosophila melanogaster*. Proc. Natl. Acad. Sci. USA 108, 9922-6. PMC3116428

Findlay, G.D., Sitnik, J.L., Wang, W., Aquadro, C.F., Clark, N.L. and Wolfner, M.F. (2014) Evolutionary rate covariation identifies new members of a protein network required for *Drosophila melanogaster* female post-mating responses. PLoS Genetics 10:e1004108. PMC3894160

Wigby, S.*, Brown, N.C., Allen, S.E., Misra, S., Sepil, I., Sitnik, J.L., Clark, A.G.*, and Wolfner, M.F.* (2020) The *Drosophila* seminal proteome and its role in sperm competition. Phil. Trans. Roy. Soc., 375(1813):20200072.

5. Triggers for egg activation in *Drosophila*

Egg activation is the process by which a mature differentiated oocyte transitions to a totipotent zygote upon fertilization; it involves meiosis completion, and modifications to maternal molecules in the egg and to the egg's envelopes. The trigger for this process in mammals is an oscillating calcium rise that is induced by the fertilizing sperm, IP3-mediated calcium release from stores, and TRPM7+CaV3.2-mediated influx from outside. We showed that, as in mammals, calcium levels rise in *Drosophila* eggs during egg activation. This rise is also due to calcium influx through a mechanically gated TRP-family channel (the TRPM7 ortholog), but in *Drosophila* the channel's activity is triggered during ovulation. As in mammals, the calcium rise initiates a calcium wave due to IP3-mediated calcium release from intracellular stores. Exploiting the large size of the *Drosophila* oocyte, we showed that egg activation involves major changes to the phospho-proteome (now also known to occur in vertebrates). The regulated proteins are products of conserved genes that our genetic tests showed to be essential for oogenesis and/or early embryogenesis. We recently connected the phosphoproteome changes to the calcium rise by showing that some of them (in cell cycle regulators, translation regulators, and others) require activity of the calcium-regulated phosphatase calcineurin, which is needed for egg activation in flies and vertebrates. Beyond our research, I chaired the 2017 GRC on Fertilization and Activation.

Horner, V.L., Czank, A., Jang, J.K., Singh, N., Williams, B.C., Puro, J., Kubli, E., Hanes, S.D., McKim, K.,S., Wolfner, M.F.* and Goldberg, M.L.* (2006) Calcium regulation by the *Drosophila* calcipressin homolog, *sarah*, is required for several aspects of egg activation. Current Biology, 16, 1441-6. (free full text)

Kaneuchi, T., Sartain, C.V., Takeo, S., Horner, V.L., Buehner, N.A., Aigaki, T.*, Wolfner, M.F.* (2015) Converging calcium waves occur as Drosophila oocytes activate. Proc. Natl. Acad. Sci. USA, 112, 791-6. PMC4311822 [*=equal, co-corresponding authors, whereever it is listed in this Biosketch.]

Zhang, Z., Ahmed-Braimah, Y.H., Goldberg, M.L., and Wolfner, M.F. (2019) Calcineurin dependent protein phosphorylation changes during egg activation in *Drosophila melanogaster*. Mol. Cell. Proteomics 18: S145-S158. PMC6427240

Hu, Q. and Wolfner, M.F. (2019) *Drosophila* Trpm mediates calcium influx during egg activation. Proc. Natl. Acad. Sci. USA 116:18994-19000. PMC6754564.

Complete List of Published Work in MyBibliography:
http://www.ncbi.nlm.nih.gov/sites/myncbi/mariana.wolfner.1/bibliograpahy/40548464/public/?sort=date&direction=ascending

PHS 398 Cover Page Supplement

OMB Number: 0925-0001

Expiration Date: 02/28/2023

1. Vertebrate Animals Section

Are vertebrate animals euthanized? ○ Yes ● No

If "Yes" to euthanasia

Is the method consistent with American Veterinary Medical Association (AVMA) guidelines?

○ Yes ○ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

○ Yes ● No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period	*Anticipated Amount ($)	*Source(s)

PHS 398 Cover Page Supplement

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ❍ Yes ● No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: http://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, check the box indicating that one from the registry will be used:

❑ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

Cell Line(s) (Example: 0004):

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ❍ Yes ● No

If "yes" then provide the HFT Compliance Assurance

If "yes" then provide the HFT Sample IRB Consent Form

5. Inventions and Patents Section (Renewal applications)

*Inventions and Patents: ❍ Yes ● No

If the answer is "Yes" then please answer the following:

*Previously Reported: ❍ Yes ❍ No

6. Change of Investigator/Change of Institution Section

❑ Change of Project Director/Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

❑ Change of Grantee Institution

*Name of former institution:

PHS 398 Modular Budget

OMB Number: 0925-0001
Expiration Date: 02/28/2023

Budget Period: 1

Start Date: 05/01/2022 End Date: 04/30/2023

A. Direct Costs

	Funds Requested ($)
Direct Cost less Consortium Indirect (F&A)*	250,000.00
Consortium Indirect (F&A)	
Total Direct Costs*	250,000.00

B. Indirect (F&A) Costs

	Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base ($)	Funds Requested ($)
1.	MTDC	64.00	235,770.00	150,893.00
2.				
3.				
4.				

Cognizant Agency (Agency Name, POC Name and Phone Number): HHS, Michael Leonard, 212-264-2069

Indirect (F&A) Rate Agreement Date

Total Indirect (F&A) Costs: 150,893.00

C. Total Direct and Indirect (F&A) Costs (A + B)

Funds Requested ($): 400,893.00

PHS 398 Modular Budget

Budget Period: 2

Start Date: 05/01/2023 End Date: 04/30/2024

A. Direct Costs

	Funds Requested ($)
Direct Cost less Consortium Indirect (F&A)*	250,000.00
Consortium Indirect (F&A)	
Total Direct Costs*	250,000.00

B. Indirect (F&A) Costs

	Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base ($)	Funds Requested ($)
1.	MTDC	64.00	235,310.00	150,598.00
2.				
3.				
4.				

Cognizant Agency (Agency Name, POC Name and Phone Number): HHS, Michael Leonard, 212-264-2069

Indirect (F&A) Rate Agreement Date

Total Indirect (F&A) Costs: 150,598.00

C. Total Direct and Indirect (F&A) Costs (A + B)

Funds Requested ($): 400,598.00

PHS 398 Modular Budget

Budget Period: 3			
Start Date: 05/01/2024 End Date: 04/30/2025			
A. Direct Costs			Funds Requested ($)
		Direct Cost less Consortium Indirect (F&A)*	250,000.00
		Consortium Indirect (F&A)	
		Total Direct Costs*	250,000.00

B. Indirect (F&A) Costs

	Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base ($)	Funds Requested ($)
1.	MTDC	64.00	234,795.00	150,269.00
2.				
3.				
4.				

Cognizant Agency (Agency Name, POC Name and Phone Number): HHS, Michael Leonard, 212-264-2069

Indirect (F&A) Rate Agreement Date

Total Indirect (F&A) Costs: 150,269.00

C. Total Direct and Indirect (F&A) Costs (A + B)	Funds Requested ($)	400,269.00

PHS 398 Modular Budget

Budget Period: 4			
Start Date: 05/01/2025 End Date: 04/30/2026			
A. Direct Costs			Funds Requested ($)
		Direct Cost less Consortium Indirect (F&A)*	250,000.00
		Consortium Indirect (F&A)	
		Total Direct Costs*	250,000.00
B. Indirect (F&A) Costs			
Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base ($)	Funds Requested ($)
1. MTDC	64.00	234,219.00	149,900.00
2.			
3.			
4.			
Cognizant Agency (Agency Name, POC Name and Phone Number)	HHS, Michael Leonard, 212-264-2069		
Indirect (F&A) Rate Agreement Date		Total Indirect (F&A) Costs	149,900.00
C. Total Direct and Indirect (F&A) Costs (A + B)		Funds Requested ($)	399,900.00

PHS 398 Modular Budget

Budget Period: 5			
Start Date: 05/01/2026 End Date: 04/30/2027			
A. Direct Costs			Funds Requested ($)
		Direct Cost less Consortium Indirect (F&A)*	250,000.00
		Consortium Indirect (F&A)	
		Total Direct Costs*	250,000.00

B. Indirect (F&A) Costs

	Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base ($)	Funds Requested ($)
1.	MTDC	64.00	233,573.00	149,487.00
2.				
3.				
4.				

Cognizant Agency (Agency Name, POC Name and Phone Number): HHS, Michael Leonard, 212-264-2069

Indirect (F&A) Rate Agreement Date

Total Indirect (F&A) Costs: 149,487.00

C. Total Direct and Indirect (F&A) Costs (A + B)	Funds Requested ($)	399,487.00

PHS 398 Modular Budget

Cumulative Budget Information

1. Total Costs, Entire Project Period

Section A, Total Direct Cost less Consortium Indirect (F&A) for Entire Project Period ($)	1,250,000.00
Section A, Total Consortium Indirect (F&A) for Entire Project Period ($)	0.00
Section A, Total Direct Costs for Entire Project Period ($)	1,250,000.00
Section B, Total Indirect (F&A) Costs for Entire Project Period ($)	751,147.00
Section C, Total Direct and Indirect (F&A) Costs (A+B) for Entire Project Period ($)	2,001,147.00

2. Budget Justifications

Personnel Justification	Personnel_Justification-ClarkWolfner.pdf
Consortium Justification	
Additional Narrative Justification	

BUDGET JUSTIFICATION

SENIOR PERSONNEL

Andrew G. Clark, PhD, **Principal Investigator** **(0.48 Academic Month)**
Dr. Clark will work closely with Dr. Wolfner on all aims of this proposed research. His particular strength and interest in evolutionary analysis of genomic data will lead him to spearhead Aim 3, including the analysis, interpretation, and paper-writing. His extensive experience in functional genomics mean that he will work closely with Dr. Wolfner in guiding the staff and trainees in the work in Aims 1 and 2, planning and critiquing the research and its results, and assisting in the writing of papers. The budget does not request salary support for Dr. Clark as his academic salary is fully supported by the University.

Mariana F. Wolfner, Co-Principal Investigator **(0.84 Academic Month)**
Dr. Wolfner will apply her experience in insect reproduction and its evolution, physiology, and molecular biology to Aims 1 and 2. Working closely with Dr. Clark and the individuals below, she will oversee and plan the research, interpret data, and write papers on this work. She will also consult with Dr. Clark on Aim 3. She will carry out all needed administrative, certification etc. tasks. No salary support requested as her salary is fully supported by the University.

OTHER PERSONNEL

Sofie Delbare, PhD, Post-doctoral Associate **(6 Calendar Months)**
Dr. Delbare has been a postdoc with Dr. Clark since 2020 (after completing thesis work with Drs. Clark and Wolfner). As a postdoc, she discovered the alternative splicing/promoter-use phenomenon that forms the basis for Aim 2. She will spend 50% of her effort on the proposed research, assisted by Ms. Jain, with the remainder of her effort supported by other sources as she works with Dr. Clark on fine-scale time courses of transcriptome change. Most of her effort will be on Aim 2, but she will also participate in Aim 3, given her interest in further statistical training.

Jolie Carlisle, PhD, Post-doctoral Associate **(6 Calendar Months)**
Dr. Carlisle, who will receive her PhD for work with Dr. Willie Swanson on the biochemical nature and evolution of abalone sperm-egg fusion proteins, will join us at the start of the proposed funding period. She will work with Dr. Delbare to carry out Aim 2, acquiring expertise with analysis of transcriptomes, RNA splicing, and gene variation. She will work 50% on this project, assisted by Ms. Jain, with the remainder of her effort supported by other sources, as she works with Dr. Wolfner on the effects of CRISPR-based substitutions of Drosophila reproductive proteins between species.

Norene Buehner, Research Support Specialist **(6.0 Calendar Months Yrs 1-3, 5.4 Yr 4-5)**
Ms. Buehner has been the Wolfner lab's Research Support Specialist for 20 years. She is highly skilled at work with Drosophila reproduction and physiology, as well as with proteins, nucleic acids and antibodies, at fly genetics (including generating and phenotyping RNAi and CRISPR'd strains), and at microscopy. She will work with Ms. Brown on Aim 1. Part of her time will continue to be spent on support duties such as ordering supplies for this project, ensuring lab-safety, and mailing out flies and lab-generated reagents (DNAs, etc.) in response to requests.

Asha Jain, Technician **(6.0 Calendar Months Yr 1-**

4, 5.4 Yr5)
Ms. Jain has been a skilled, thoughtful, hardworking and reliable technician in the Clark lab since Aug 2018. She is skilled at all aspects of the front end of genomics technologies, including basic molecular biology and library preps of all sorts. She also has experience with all aspects of Drosophila handling, including gonadal dissections. She will assist Drs. Delbare and Carlisle with Aim 2.

Nora Brown, Graduate Student (12.0 Calendar Months)
Ms. Brown is an evolutionarily-focused student in our graduate program in Genetics, Genomics, and Development, whom we co-mentor. She will carry out Aim 1, assisted by Ms. Buehner. She is well qualified for the proposed experiments given her experience with Drosophila reproductive biology and with in silico analyses of reproductive genes and their evolution, and her excitement about fly mating plugs stemming from her discovery that a component of this plug has taken on this expression in some but not all Drosophila species. Given her interest in statistics (including TAing statistics a year ago), she will also participate in the analyses in Aim 3, as time permits.

TBN, Undergraduate Students (2 in yrs 1-2, 1 in yrs 3-5) (2.5 Summer Months each)
The Drosophila studies often benefit from having well-supervised undergraduates assist in the more labor-intensive aspects. These students learn fundamental lab skills, and several have produced excellent theses. For example, the initial GWAS of mating plug ejection was largely scored by undergraduates.

PHS 398 Research Plan

OMB Number: 0925-0001
Expiration Date: 02/28/2023

Introduction	
1. Introduction to Application (for Resubmission and Revision applications)	
Research Plan Section	
2. Specific Aims	SpecificAims-ClarkWolfner.pdf
3. Research Strategy*	ResearchStrategyClarkWolfner.pdf
4. Progress Report Publication List	Progress_report_papers-ClarkWolfner.pdf
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	MultiPILeadPlan-ClarkWolfner.pdf
8. Consortium/Contractual Arrangements	
9. Letters of Support	Marla_Support_Letter.pdf
10. Resource Sharing Plan(s)	ResourceSharingPlan-ClarkWolfner.pdf
11. Authentication of Key Biological and/or Chemical Resources	AuthenKeyBiolChemRes.pdf
Appendix	
12. Appendix	

SPECIFIC AIMS

Genetic variation in both males and females has a profound effect on the successful production of healthy zygotes. This variation mediates a complex set of interactions between the sexes at the molecular, cellular, neuronal and whole-organism levels. Our past work has focused on the genetic and molecular factors important for differential success of the sperm in multiply-mated female Drosophila. We identified high levels of naturally occurring genetic variation in male- or female-expressed genes that impact this sperm competition, including genes that affect neuronal function in the female. In this renewal application, we will dissect the roles of these genes in the differential timing of mating plug ejection, differential alternative splicing in the female and its modulation by seminal fluid components, and we will apply a novel modeling approach that allows quantitative assessment of the rank ordering and transitivity of components of sperm competitive success.

Specific Aim 1: Identify genetic determinants of mating plug ejection timing. When a female Drosophila mates, she has time to assess the quality of the male and his ejaculate and to modulate the likelihood that his sperm will be advantaged to fertilize her eggs. For example, females vary in the timing with which they eject the mating plug that formed during copulation. If ejection is delayed, more sperm are able to go into storage and be used for subsequent fertilizations, whereas early ejection implies greater loss of that male's sperm. Because mating plug ejection timing plays such a vital role in determining the outcome of sperm competition, we began a GWA study, scoring ejection-timing across DGRP lines to identify genes in females that might play a role. Preliminary data show extensive variation in mating plug ejection timing among female genotypes, with some interesting candidate variants driving the differences. In particular, octopamine (OA) neurons innervate the female reproductive tract, and our preliminary GWAS identified an OA receptor; other neural genes also show association. We propose to complete this GWAS, and perform a grid cross to identify both male and female genes that impact the female's decision when to perform mating plug ejection, and follow up with validations and tests of interactions between the two.

Specific Aim 2: Characterize and quantify alternative splicing/promoter use as a key component of female response to mating. Careful examination of the transcripts produced by the female brain in a detailed time series after mating has identified a large number of genes that display mating-dependent differential alternative splicing, including genes involved in seeking or pheromone-related behaviors. This implies that molecules in the male ejaculate, including one that we pinpointed, manipulate the female's genome not only by the previously-known effects on transcript levels, but also by directing differential promoter use and/or production of alternative gene splice products. This gives rise to a slew of questions – Does this impact the reproductive success of the female and her ability to discriminate and decide between males? Does it provide a reproductive advantage to a given male? Is there functional variation in the degree of alternative splicing directed by the male? Our experiments will determine when and how molecular signals from the male drive a program of alternative splicing/promoter use in the female, and whether these changes impact male reproductive fitness, female reproductive fitness or both. Finally, we will assess in collections of reference lines of *D. melanogaster* the degree to which variation in sperm competition outcomes are associated with variation in alternative splicing/promoter-use, and using association mapping methods, identify genes that modulate variation in mating-induced alternative splicing and alternative promoter usage.

Specific Aim 3: Apply Bradley-Terry models to assess the nature of pairwise interactions in mating. Reproduction seems like such a multifactorial and complex phenomenon that prediction of success must surely be entangled in so many interactions as to be impossible. But not all phenomena that start inherently with a pairwise interaction defy prediction. In sperm competition, we frequently want to know whether, when genotype A beats genotype B, and B beats C, does this mean that A beats C? This kind of question about transitivity can be taken a step further to assess whether we can predict, from the other pairwise contests, by how much A beats C. Modeling this sort of situation, with many pairwise contests examined among a set of mating partners (or players, or teams) arises in many contexts, and there are models that statistically fit the data to quantify the rank order of the "quality" of each line, and render such predictions. Existence of such predictions, if they are accurate, implies that the process is not entangled in an impenetrably complex web of interactions, but instead the many factors that impact the outcome boil down to a rank ordering of male quality and female quality. We propose to apply models of this sort to not only sperm competition data, but also to the mating plug timing data of Aim 1, and the alternative splicing/promoter-use data of Aim 2. Fits of these models will address a fundamental question about the level of complexity that is faced in predicting sperm competition outcomes.

Many of the processes that we study here show high levels of evolutionary conservation, implying that our results will expand the foundation for understanding male x female interactions in reproduction that may have relevance to cases of idiopathic human infertility that may involve genetic incompatibility between the partners.

RESEARCH STRATEGY

While males and females share a common interest in producing healthy offspring, the actions of both sexual selection and natural selection often promote genes whose effects are favorable in one sex and disfavorable in the other. We seek to elucidate mechanisms for these sexual conflicts. Most studies that address the evolution and functions of these genes have tended to focus on one sex or the other, when in fact the interaction between the sexes is what drives the evolution of these traits. Previous work has established sperm-competition/female sperm-preference (we will use 'sperm competition' as shorthand, here) as an ideal system in which to establish the evolutionary consequences of male- and female- genotypes simultaneously: for example, the outcome of sperm competition between two males depends on the genotype of the female with which they mate. Thus, neither male nor female can evolve unconstrained by the genotype of their partner.

Under the auspices of this grant, we examine these male-by-female interactions by using genetic variation in the Drosophila model system. We identify critical genes, alleles, and processes and then test how they act in sperm competition. Most research on sperm competition has focused on the male side, showing that sperm number and quality, and seminal proteins, influence the outcome of this process (3, 4), which is thought to have driven selection for large numbers of sperm and certain activities of seminal proteins (like making insect females reluctant to remate). During the present funding period we examined another potential male contribution – the exosomes that are transferred to females along with sperm. These small vesicles accompany sperm into females in mammals and flies (5, 6), and have been shown or suggested to modify either sperm or the female's reproductive tract to improve fertility outcomes. Drosophila exosomes produced in the male's accessory gland have been shown by the Wilson lab to affect the female's post-mating behavioral response (6). During this funding period we discovered variation in these exosome numbers across the Drosophila Genetic Reference Panel (DGRP; a panel of strains with different wild-derived genomes (7) (Fig. 1); we plan to complete this analysis and its intersection with sperm competition outcomes by the end of the present funding period. We have focused much of our effort on our initial studies on the more elusive female side of the equation.

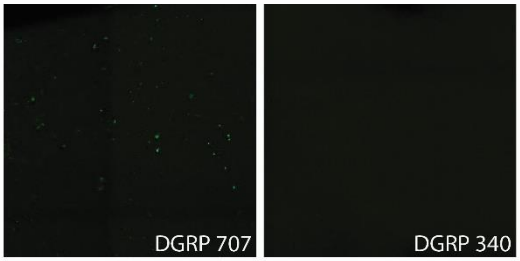


Fig. 1 Anti-chaoptin labeling (green puncta) of exosomes in comparable regions of the lumen of male accessory glands of two DGRP lines, line 707 with many exosomes and line 304 with nearly none.

Exploring the female determinants of sperm competition outcomes, we performed a genome-wide association study (GWAS) on the DGRP strains (many distinct female DGRP strains doubly mated with the same pair of male strains), followed by RNAi analysis in sperm competition assays. We identified 8 genes whose activity in females regulated sperm precedence. Two were expressed in sperm-storage organs, a logical place for a sperm-competition influencing gene product. Interestingly, the other 6 genes were expressed in the nervous system, implicating an active role by the female in the process of sperm competition/preference. There are a variety of ways in which a female's nervous system could impact the outcome of sperm competition. First, the female could actively control the fate of the sperm she received, such as by regulating muscular contractions in her reproductive tract to either release or retain sperm preferentially from a particular male. We will investigate this aspect in Aim 1. Second, the female's nervous system could be essential for her sensing the quality of a particular male, sensing her metabolic or mating status, or for adapting her behaviors accordingly. Recently, for example, it has been reported that female Drosophila become more "choosy" about mates after a first-mating (8, 9), thus changing the stringency of the criteria with which they evaluate a subsequent male. In Aim 2, we will investigate the nature, variation, and consequences of mating-induced changes in alternative splicing and promoter use in the female brain, which have the potential to alter such behaviors. Finally, in the third Aim, we will develop a theoretical model to see which of the variable effects we have dissected previously or during the proposed funding period play the largest roles in predicting the outcomes of sperm competition.

Further motivation for the work we propose here comes from progress in the present funding period in several related areas. We discovered variation in transcriptional responses by females after mating to males from different *D. melanogaster* populations (10, 11) and also as a consequence of the state of their gut microbiome (12).These findings and expertise also led to the results that motivate proposed Aim 2. We developed a method to test the genes that we identified early in the current funding-period (13) for functions in heterospecific sperm competition situations involving a *D. simulans* male. Specifically, using a new *ppk23* mutant strain of *D. simulans* (14), we were able to get better cross-species mating, although it was still difficult to get sufficient matings for robust sample-sizes. Then we realized that the *D. americana* x *D. novamexicana* species pair readily interbreed, and thus can serve as an ideal model for hetero- vs. conspecific mating

reactions. With this model, we showed that females react rapidly to heterospecific matings with a transcriptional stress response, activating genes in the Imd and JAK/STAT pathways (15). Exploring the male-side of sperm competition, we identified genes expressed in secondary cells of the male's accessory gland that regulate his sperm competitive success. Intriguingly, one of these genes encodes a non-coding RNA and another encodes a micropeptide, one of the first of the latter to be assigned a biological function in Drosophila (16-18). We advanced our understanding of variation in neuronal genes on the female's response to mating, elucidating an effect of allelic variants in "Tdc2" octopaminergic neurons in the female's ability to discriminate among ejaculates (13). In addition, we investigated or reviewed multiple facets of male x female interactions that are important in sperm competition/preference outcomes including behaviors ((19-23)), seminal proteins (3, 24) (25), octopamine (26), the microbiome (12), and the Y chromosome (27). Finally, we validated the relevance of our DGRP/RNAi approach to questions beyond Drosophila, such as for identifying genes of relevance to human diseases (28, 29).

1. Significance

By testing whether mating plug ejection is the female behavior that best predicts sperm competition outcomes, we will be able to focus attention on neuronal pathways and their molecular inputs that drive the timing of this behavior (Aim 1). The directness of the link between gene action and this female behavior will be a significant advance in teasing apart determinants of sperm competition outcomes. We will elucidate male traits that respond to this, and the extent of male x female interactions in this process. Our results may also shed light on new roles for semen coagulation in humans and other mammals, and copulatory plugs in the latter.

The radical changes in alternative exon usage that we saw in post-mating females open the opportunity to discover genetic variants (and their pathways) that drive this impressive molecular response (Aim 2). We argue that this may be an important way that males can evade a female counter-response to his manipulation of her physiology and transcriptome. The many post-mating changes in use of alternative promoters and alternative splices raises the questions: 1) what drives these changes, 2) is variation in sperm usage caused by variation in alternative transcript use, and 3) why did evolution choose this route to regulate mating responses? The answers will advance our understanding of post-mating responses, and how they impact the reproductive success of pairings, and how they vary in advantaging particular males in the context of particular females.

Models that assess attributes of males and females in a way that allows prediction of novel pairwise combinations are central to the theme of Aim 3 of this grant. Here we propose to develop an approach to modeling sperm competition that uniquely accommodates the fact that its study is done through large series of pairwise tests. The challenge to predict outcomes of pairwise contests, even when particular pairs have not previously encountered one another, arises in many contexts, and so this theory has deep roots (30). It has appeal here because the demonstration of predictability of outcomes implies something fundamental about the system, namely that the rank-ordering of male and female effects (e.g. on sperm competition outcomes) is sufficient. Do some processes (e.g. specific transcript abundances, or mating plug ejection) fit the model perfectly, whereas others do not? Rejection of models is often more informative than failure-to-reject, and this may well be the case here. We will apply tests in many contexts, from sperm competition itself, to many of its component parts (e.g. mating plug ejection timing, single gene transcript levels, and alternative promoter/exon use). The same paradigm can be used to model many other aspects of reproductive biology, realizing that sexual reproduction is the prime "pairwise contest" in all of biology.

2. Innovation

Each of the three specific Aims of this proposal poses a novel and innovative question that has consequences for both the functional coordination as well as the evolution of male and female behavior, physiology, and biochemistry to successfully produce fertilized eggs and viable offspring. First, while current data suggest that mating plug ejection timing is an important contributor to sperm competition outcomes, this has never been tested systematically. By using scaled-up videography to efficiently score and quantify the timing of mating plug ejection, Aim 1 will directly determine how well mating plug ejection timing correlates with sperm competition outcomes. Sensing the quality of the male and his ejaculate and deciding when to eject the mating plug is the most direct, proximal action by which females modulate sperm competition outcomes. The basis of this critical phenomenon is completely unknown. By applying our approach of identifying genetic variants associated with differences in mating plug ejection timing, we expect to learn if the variation arises due to differences in sensing, perception, processing by the female, or the physical properties of the mating plug itself, or the mechanics of its removal by the female.

While alternative splicing and alternative promoter use are known to be important in diversifying the proteome and gene expression regulation in many contexts (31-33), to our knowledge these aspects have never been studied in a reproductive context. Aim 2's proposed investigation of alternative exon-use induced in brains by mating is the first step toward dissecting this new area. In addition to elucidating mechanisms important to sperm competition, the results will likely bring a new perspective to the understanding of mate-sensing behavior, which has previously been examined only in non-competitive situations. Dissecting the new phenomenon of control of a female's alternative splicing/promoter use by a male raises questions of why would evolution would produce this outcome? How do males manipulate the female by modulating her transcript usage? We posit that modulation of transcript usage might be more difficult for the female to counter compared to total gene expression changes.

The methods in Aim 3, developing models for analysis of pairwise contests, will be new and predictive and have also relevance to many aspects of reproductive biology. Prediction is not something that you encounter much in the field of sperm competition, but ability to predict outcomes is one important test for how well we understand a phenomenon. By quantifying the degree to which males can be rank-ordered in their ejaculate qualities, and females can be rank-ordered in the proclivity to assess males' qualities, we can finally test for a large number of attributes for which ones drive the best predictions. Finally, we emphasize the long and successful track record of our unique collaboration, bringing together a synergistic blend of rigorous Drosophila genetics with evolutionary, population genetic, molecular, statistical and computational approaches to solve exciting challenges in reproductive biology.

3. Approach

Aim 1: Identify genetic determinants of mating plug ejection timing.

The involvement of the female's neurons in the relative progeny outcomes of a double-mating could be due to proximal effects controlling the storage and release of sperm and/or to behavioral or other effects. During the present funding period, we determined that the octopaminergic neural system is essential for the female's regulation of sperm precedence (13, 34). Our knockdowns of genes from our GWAS specifically in the octopaminergic Tdc2 neurons altered sperm competition outcomes (35), clearly implicating this neural system in the female's treatment of alternative ejaculates. Tdc2 neurons, which make the neuromodulators octopamine (OA) and its precursor tyramine (TA) regulate a number of important processes in flies (reviewed in (26)). Of particular relevance to this proposal, these neurons innervate the female's reproductive tract and OA regulates muscle contractions there, such as those that allow ovulation (36). Moreover OA (and TA) regulates the rate of sperm release from storage (37), and Tdc2 neurons innervate the base of the sperm-storage organs (37). We found that sperm competition outcomes are altered in mutant females who lack OA: the first male mated to such females does better than he would in a competitive situation (higher P1) relative to the P1 outcome had he mated to a control female. By disabling OA neurons specifically (with UAS-TNT), we confirmed that these neurons' activity is essential for sperm competition outcome (Fig. 2): Tdc2-GAL4>UAS-TNT females produced a higher proportion of first-male progeny than controls (38). Further dissection of sperm-handling events in doubly-mated females suggest a model in which female OA is needed not for differential storage of either males' sperm, but instead to preferentially use second-male sperm for fertilization; thus, absence of OA, or disabling OA neurons, would result in a higher P1. In this Aim, we investigate a likely candidate for these local effects on sperm handling by reproductive tract muscular processes that control the outcome of sperm competition (in Aim 2, we examine the brain, likely locus for behavioral effects). Specifically, we will determine genes that regulate the timing of ejection of sperm by the female, and effects of their variants on sperm competition outcomes.

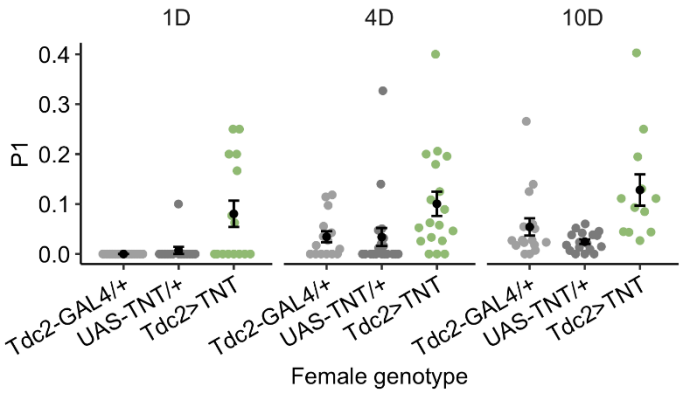


Fig. 2 Ablation of Tdc2 neurons (the Tdc2>TNT females; green) results in greater first-male paternity (P1) compared to Tdc2-GAL4 or UAS-TNT sibling controls (grey). This is consistently true at 1 day, 4 days and 10 days of progeny collection.

Aim 1a: What are the genetic determinants of variation in mating plug ejection time in females?

Controlling release of sperm from storage (as with OA, above) is but one way that females can manipulate sperm use. Another is to control the number of sperm she stores from a particular male. Sperm and seminal proteins coagulate within the mated female's reproductive tract to form a "mating plug" (39, 40). The mating plug, which only forms within the female and contains female as well as male components (39, 40), retains sperm in the female, allowing them to be stored. Early ejection of the mating plug prevents retention of sperm, resulting in fewer being stored; retention of the mating plug provides more opportunities for sperm to be stored; in multiply mated females ejection timing influences the paternity share of her mates (39-42). Thus, the mating plug – and the female's control of its ejection – is a prime locus for male x female interaction that can affect sperm competition outcomes. In one example, *D. simulans* females previously mated to conspecific males eject the mating plug much sooner if they remate with a heterospecific *mauritiana* male than with a conspecific male (43). In another example, in interspecific matings of cactophilic Drosophila species a solid hard "insemination reaction mass" forms in place of a normal mating plug, preventing sperm from being stored in the female (44). The mating plug also helps initially retain seminal proteins in the female (39) and contains pheromones that repel other males (45), both aiding the reproductive success of the focal male.

Female control of the timing of mating plug ejection provides a way for her to regulate the amount of sperm she stores from a given male. The diuretic hormone 44 (*Dh44*) neural circuit influences mating plug ejection timing: experimentally activating *Dh44*$^{+}$ neurons delays ejection, while silencing these neurons expedites ejection (42). However, beyond involvement of these neurons, nothing is known about the control of mating plug ejection. To identify female actors in this mating plug ejection, we tested for genetic variation in mating plug ejection across a small sample of Drosophila Genetic Reference Panel (DGRP) lines (7). In preliminary experiments we tested the timing of mating plug ejection by females from 30 DGRP lines. We mated 30 females from each line, individually, with single Canton S males. After placing just-mated females into observation chambers (46), we determined, the time of ejection of the mating plug (easily visualized due to the plug's fluorescence, because of its fluorescent constituent protein pEBme (47)). We observed significant variation in mating plug ejection times among the lines (Fig. 3), indicating a clear genetic component to the female's control of mating plug ejection times. Recognizing the statistical uncertainty of a GWAS with only 30 lines, we carried out a preliminary screen for variants associated with in mating plug ejection timings, and were excited to find an octopamine receptor (*Oct-beta-3R*) and a neural and reproductive tract gene (*trol*) among the top 15 hits.

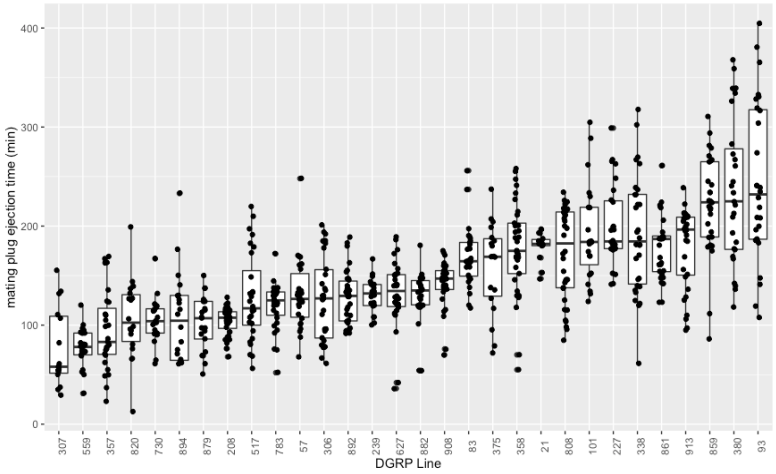


Fig. 3 Time post-mating of ejection of mating plug for ~20 females from each of 30 DGRP lines. The line means are significantly different ($P < 2.2 \times 10^{-16}$, ANOVA).

The first task in Aim 1a is to complete the GWAS to identify candidate genes for female regulators of mating plug ejection, and then to test each candidate's roles in mating plug ejection. This will require tests of mating plug ejection time in an additional 30 DGRP lines to have enough for a robust GWAS. This round will be dramatically accelerated by use of a 24-well chamber for the females combined with simple digital videography. By simply scrolling through mpeg files for the first appearance of the fluorescent plug, we can quickly score the time of each plug ejection. We will then test the function of each GWAS candidate in mating plug ejection genetically. We will obtain from the Bloomington Drosophila Stock Center (BDSC) or generate ourselves, using CRISPR/Cas9, a null mutant in each candidate gene. We will focus on the top 10-20 candidates, prioritizing those with neural or reproductive tract expression, where known. If the mutant is viable, we will test mutant females for their timing of mating-plug ejection after mating with our standard male, using the assay described above. If mutants of a candidate gene are not viable, we will use RNAi to conditionally knock down the gene in adults (with *Tubulin*-GAL4; *Gal80*[ts] (48)) or locally, where appropriate (such as with a pan-neuronal (*elav*-GAL4 (49)), neural (*Dh44R1*-GAL4 (42) (and *Tdc2*-GAL4 (50) if results in Aim 1c, below, indicate involvement of these neurons)) or reproductive tract (Oamb-GAL4 (51)) driver]). After knockdown is verified, we will test knockdown females for mating plug ejection times after mating with Canton S males as above. Mutants or knockdowns that show significantly different times of mating plug ejection relative to

genetically matched controls will be kept for further study. Our results will identify genes that control mating plug ejection; studies with RNAi or local knockouts (52) will then determine where their action is needed.

Aim 1b: What is the relationship between mating plug ejection speed and sperm competition outcome?

We previously typed nearly 40 DGRP lines for male- and female-based variation in sperm competition outcomes. In this sub-aim, we will determine whether their sperm competition variation correlates with variation in mating plug ejection timing by females. For example, to examine the male-contribution to the interaction, using ten of the DGRP lines whose males we found to have a wide range of P1 and P2 values (53), we will set up single pair crosses with Canton-S females, isolate females when mating is complete, and score mating plug ejection timing by videography, as described above. The females will then be placed into vials to collect offspring from the first mating. Two days later, each female (each tracked individually) will be given two bw^D males for a second mating; relative numbers of non-bw^D progeny from each female will provide the P1 score for the DGRP males. After each female's second mating, we will score mating plug ejection timing again, and the females will be transferred to vials to collect progeny. Thus, this experiment will measure the ejection timing for both mating 1 and 2, as well as the sperm competition parameter P1. With 30 replicates for each male, this will be well powered to detect a correlation between P1 and mating plug ejection timings; it will also allow us to determine whether females adjust ejection timing for first vs. second matings. The mirror image of this entire experiment will be repeated, using bw^D males for the first mating, and the DGRP males for the second, and in this case we can quantify the correlation between mating plug ejection timings and the P2 sperm competition parameter. Note that we can also partition the covariance between the two into a within-line and between-line component. To examine the female-contribution to the interaction, we will take females from 10 DGRP lines that yielded a wide range of mating plug ejection timings (above) or P1 and P2 values with standard males (34), and will mate them to those two standard, genetically distinguishable males (*cn bw*, and then bw^D, analogous to the procedure just above. Here too we will measure variation in mating plug ejection timing after each mating, and test whether either or both correlate with sperm competition outcomes. If the variation in mating plug ejection timing is correlated with the variation in sperm competition/preference outcomes, it suggests that variation in mating plug ejection timing is a major contributor to sperm competition/precedence, and will tell us whether these are female-, male- or both-driven. We strongly expect to see such a correlation, given the reports on Dh44 and interspecies experiments noted earlier (42, 43), but if we do not see a correlation it would indicate that while mating plug ejection was critical for sperm competition outcomes in those studies, it is not the main basis for the effect seen across the DGRP; this in itself would be interesting, suggesting an unusually high flexibility in sperm competition mechanisms.

Aim 1c: What is the role of octopamine in the regulation of mating plug ejection?

Since we have strongly implicated Tdc2 neurons and OA signaling in sperm competition, it is logical to ask whether the mechanism whereby OA exerts its influence of sperm competition includes regulating mating plug ejection timing. The GWAS of Aim 1a may provide a tentative answer, but direct manipulation of Tdc2 neurons will be more definitive. We will examine mating plug ejection timing in females mutant for OA-generating enzymes (Tbh, Tdc2; (50, 54)) or in whom the OA-producing Tdc2 neurons were inactivated by using UAS-TNT(55) (driven by *Tdc2*-GAL4 (50)), after mating to a standard (Canton S) control male. If Tdc2 neurons regulate mating plug ejection timing, we will see a difference in timing in females with these neurons inactivated. In this case we will also test whether ectopically silencing or activating these neurons (by temperature shifts of *Tdc2*-GAL4; UAS-*shi*[ts] (56) or *Tdc2*-GAL4; UAS-*dTrpA1* (57), respectively) causes an effect on mating plug ejection as well.

Aim 1d: What is the relationship between <u>genetic variation</u> in mating plug ejection time and sperm competition performance?

If we find a correlation between the lines affecting mating plug ejection and sperm competition/preference outcomes and/or effects of OA signaling on both processes, we will test whether the 6 genes that we previously identified as regulating sperm-competition outcomes via the female's nervous system (13, 34) exert this effect by regulating the timing of mating plug ejection timing. Indeed, individual knockdown of 5 of these genes decreased P1, consistent with roles in delayed ejection timing. To test whether these genes regulate the timing of mating plug ejection, we will generate females knocked down for each of these genes, individually, with pan-neuronal, Dh44-neuron (and, if appropriate given Aim 1c, Tdc2-neuron) drivers, and determine the timing of mating plug ejection (relative to matched controls) after mating with a standard male. If this manipulation results in altered mating plug ejection timing we will know that this is one way that these genes influence this process. If not, then the gene affects other phenomena that alter sperm competition/preference outcomes, such as sperm release from storage. Conversely, we will test the genes defined in Aim 1a as critical

for controlling mating plug timing for their effects on sperm competition (by assays as in Aim 1b (13)). If a genetic variant that we identified as important in mating plug ejection timing also affects sperm competition outcomes, it suggests that its action in the latter is through its effects on mating plug ejection timing. Alternatively, if a genetic variant affects mating plug ejection timing but not sperm competition outcomes, it suggests that this gene's effects are more specific to mating plug ejection, and the sperm competition change is driven by other factors.

Aim 1e: To what degree does male genetic variation modulate a female's mating plug ejection time, and how can we quantify male x female interaction in mating plug ejection?

In our initial tests of the 30 DGRP lines, and in the sub-aims proposed above, we have examined mating plug ejection by females (DGRP, mutant, knockdown) after mating with a standard male. However, we know that sperm competition/preference is the outcome of male x female genotypic interactions, and that the mating plug contains important male contributions (e.g. (39, 47, 58)). Therefore, in this sub-aim, we will return to the DGRP to examine the 'male side' (mating plug ejection rates by a standard female after crossing to males from different DGRP lines). We will use the same lines as in Aim 1a above, with the goal of determining whether the same GWAS 'hits' (or mutants/knockdowns) participate in both male and female determinants of ejection timing. Our results will also identify, via GWAS, potential male-determinants of mating plug ejection dynamics; it is likely that at least some of those will correspond to genes already known to encode mating plug contributions from the male, or seminal proteins with roles in sperm competition (e.g. (17, 18, 39, 47, 58-61).

In order to assess male x female interactions in the determination of mating plug ejection timing, we will employ a 6x6 grid-cross design. Here we will switch to lines from the Drosophila Synthetic Population Resource DSPR lines (62), which represent global *D. melanogaster* diversity. The DSPR have excellent quality de novo PacBio genome assemblies, and a large collection of RI line descendants, subsets of which can be used to efficiently map differences between any given pair of lines. After each mating, females will be followed as was done for our GWA study, scoring the mating plug ejection times. Data from this experiment, which will be well replicated (20 females from each cell over two blocks), will provide a balanced assessment of the role of both male and female genetic variation, as well as formal tests of genetic interaction. We refer the reader to Aim 2 for a parallel study of transcript use, and Aim 3 to see application of the Bradley-Terry model for this analysis, an approach designed for such pairwise contests and uniquely suited to the question.

Potential challenges and alternative approaches

The methods for mating plug ejection assays, GWAS, RNAi, and CRISPR/Cas9 genome editing are all straightforward and in use in our lab; we do not anticipate technical challenges with these. All Drosophila stocks necessary for these studies are already in hand, including all needed GAL4 drivers. For manipulating individual genes, we expect that RNAi-generating stocks for any gene of interest will be available from the VDRC (63) or among TRiP lines (64) at the BDSC, as has been the case for any RNAi line we have needed to date; however if a line is not available we can generate it ourselves, as we did in (65). Similarly, CRISPR/Cas9 mutagenesis is something we do routinely (for example: (66)); for genes for which TKO lines (67) are available from the BDSC, generating a knockout will be even quicker (just a matter of genetic-crossing).

Although there is strong evidence from multiple groups that mating plug ejection plays an important role in sperm competition/preference outcomes (41, 42), it is possible that some genes (13) or molecules (OA) that we know to regulate sperm competition outcomes will not affect mating plug ejection. That would simply indicate that these genes/molecules act at other steps in sperm competition – a complex process that our work is the first to parse into mechanistic components. We have the assays and experience to test these genes for roles in other features of this process (e.g. movement of sperm into or out of storage, remating-receptivity, etc. (37, 65, 68, 69)). Alternately, there is a chance that some of the genes that regulate the timing of mating plug ejection may not show a dramatic effect on sperm competition/preference outcomes. This will alter the view of the field in that mating plug ejection is a central component in sperm competition/preference.

Aim 2: Characterize and quantify alternative splicing/promoter use as a key component of female response to mating.

The male achieves his objective of "winning" sperm competition not only by making "better" or more sperm or the seminal proteins that support them, but also by modifying the physiology of females to favor the couple's fertility, and hence his own fitness. Our labs have investigated the transcriptome-level interactions between male and female genotypes that could result in this manipulation. We have shown that the transcriptomes of

females are affected differently by different males with which they mate (10, 11); for example immune mediators are one type that differ in the magnitude of post-mating activation depending on the female's mate. A key finding from our genetic studies is that the nervous system of the female is critical in sperm competition outcomes (13, 34, 38). In this aim we examine male x female interactions in the transcriptome of the female's CNS/brain that could control females' abilities to sense male quality or, alternatively, to modify her behaviors after mating.

In order to precisely quantify post-mating neuronal gene expression changes in the female, we examined the head transcriptome of Canton-S females following a single mating with either wild-type or males that lack a seminal fluid peptide, Sex Peptide, that is known to regulate post-mating behavioral changes in females (2). We isolated RNA samples from 10 timepoints post-mating (from 30 min to 24 hr, controlled for the time of day), to infer gene regulatory networks from the time series data (70). We found 223 genes that were differentially expressed between any of the 3 treatments, with at least a 2-fold change in expression at least at one time point. Virtually all post-mating gene expression changes in the first 5 hr after mating occurred independently of Sex Peptide, as expected as it takes several hours for effects of Sex Peptide and other seminal proteins to be evident as these proteins need to reach, and modify, their target tissues (36, 66, 71, 72). After 5 hr, effects of Sex Peptide were evident: females mated to Sex Peptide null males had a transcriptome profile that was almost identical to that of virgin females. Sex Peptide-dependent differentially-expressed genes were significantly enriched for ones involved in ribosome biogenesis, lipid and carbohydrate metabolism, and included ones that encode transcription factors involved in neurogenesis, and odorant binding proteins.

Extending beyond changes in gene expression level, we examined read counts to quantify exon-specific RNA abundance, which allow inference of alternative splicing and alternative promoter usage. We were intrigued to find examples of differential exon use in the heads of mated vs. unmated females. Whenever the change occurred after 5 hr post-mating, those changes were induced by the seminal protein Sex Peptide (Fig. 4). A recent report on cactophilic Drosophila validates the generality of our finding (73) but those species lack the tools we have in *D. melanogaster* for analyses such as those we propose below. Such modulation of female transcript usage is an entirely new way for a male to influence his mate's responses, and our observation immediately raises questions about its molecular mechanism, as well as its variability and evolution. In *D. melanogaster* heads, we found evidence for differential exon usage in 86 exons across 74 genes (with at least a 1.5-fold difference). Only one gene was both differentially expressed and had differential exon use, highlighting the importance of looking at the exon/transcript structure effects – beyond effects on overall level of gene expression. The 74 genes with differential exon use were enriched for GO terms related to fatty acid and pheromone metabolism, neuron development and behavior. The genes include several candidates that are interesting to study for a potential role in the regulation of post-mating responses. These candidate genes include two, *foraging (for)* and *Desat1*, that are especially intriguing in light of behavioral changes in females that can affect sperm competition outcomes, and several gene involved in signaling by the steroid hormone 20-hydroxy ecdysone (20HE), which is modulated post-mating and influences some post-mating physiological responses. While the latter genes (the 20HE receptor *EcR*, and *DOR, Eip74EF* and *Eip75B*) are also intriguing, we will focus our studies in sub-Aim 2b on *for* and *Desat1* as they are linked to behavioral outcomes.

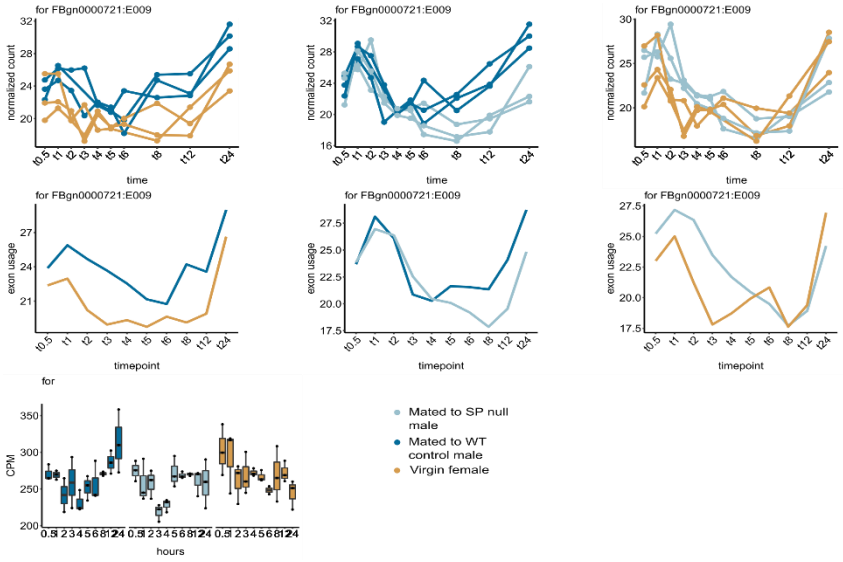


Fig. 4 Normalized read counts from foraging *exon 9 in heads of females that are virgin, mated, and mated to Sex Peptide null males. Top row shows 3 replicates of each of the 3 pairwise comparisons. Second row are treatment means over replicates. The bottom panel depicts mean CPM of foraging transcripts for the 3 treatments across time. (Data from (2)).*

Foraging affects searching behaviors, feeding, learning/memory, sleep, and other responses that contribute to active assessment of a mate (reviewed in (74)) and that are known to be regulated by Sex Peptide, although *foraging* has never been studied for a role in reproduction. Alternative promoter and splice site use in 21 *foraging* transcripts, associated with different behaviors (1) (Fig. 5). The *foraging* transcripts can encode 12 protein isoforms, all of which share a cGMP binding domain and a protein kinase domain, but likely have different substrates (74). They derive from four promoters, which are active in distinct sets of tissues, in both larvae and adult flies (1). The human homolog of *for* is *PRKG1*. Polymorphisms in *PRKG1* have been correlated with differences in "goal pursuit behaviors," interactions between lifetime trauma and alcohol abuse, and maternal sensitivity towards infants after early life adversity ((74)).

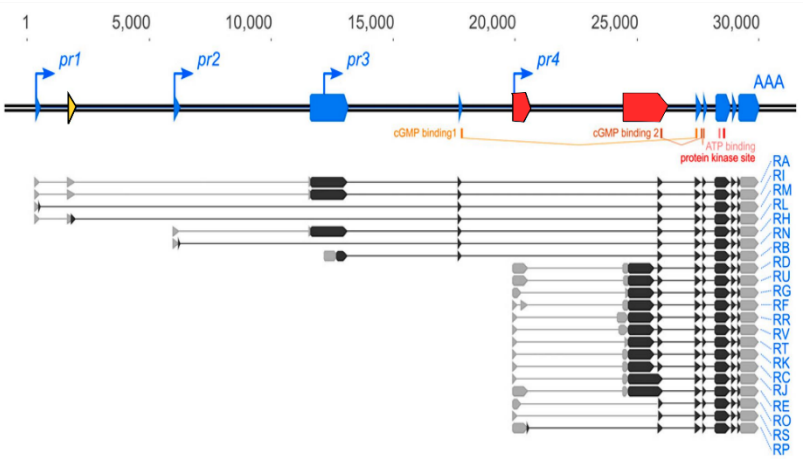


Fig. 5 Alternative transcripts of the gene foraging. *Note the presence of 4 distinct promoters. Promoter 4 (pr4) is the one that is preferentially used in post-mating females. Red exons are up-regulated in response to Sex Peptide; yellow is downregulated. (modified from (1)).*

Desaturase-1 (*Desat1*), a gene known for its role in metabolism of cuticular hydrocarbons that play a key pheromonal role in mating (75). *Desat1* has five promoters, which all give rise to the same protein but are expressed differently. For example, Bousquet et al. (76) found that the different promoters are active in distinct regions of the brain. We found that, in response to mating and Sex Peptide, there is increased use of the fifth promoter of *Desat*1 (which gives rise to transcript isoform RD). Bousquet et al. (76) found that RNAi against transcripts of the RD (fifth) promoter region led males to fail to discriminate male vs. female sex pheromones (based on observed courtship behavior). Thus, *Desat1* plays not only a role in pheromone synthesis, but also, using promoter 5, in pheromone discrimination. Thus, it is possible that the change in *Desat1* promoter use alters pheromone discrimination in mated females; it is not difficult to imagine how this may lead to altered sperm competition outcomes, as sensing of male quality appears to play a role in mating plug ejection time.

These results, together, motivate further analysis of male manipulation of alternative splicing and promoter use in the female. What is the extent of male variation in this manipulation? Are different females differentially responsive? Do genes, such as *for* and *Desat1*, that exhibit alternative exon/promoter use present hypotheses for an adaptive advantage to the male to mediate female gene expression in this way?

Aim 2a: Is variation in modulation of post-mating alternative transcript use directed primarily by the female's genotype, the male's genotype, or is it the result of male x female genotype interactions?

We know that the post-mating transcriptional response in females strongly varies with the male genotype (10). The above discussion underscores the observation that there is considerable response in females at the level of differential promoter usage and alternative splicing. The next logical question is whether genetic variation among males results in differential modulation of the female promoter usage or alternative splicing. In the same experiment we can also ask about variation in the female response and the magnitude of male x female interactions. Results of this experiment will direct our attention to the possible genes and pathways that modulate the alternative transcript usage in females.

We will determine this through a 6x6 grid cross among six parental DSPR lines, designed analogous to that described in Aim 1 for analysis of mating plug ejection. In this case, the 36 matings will be set up, and females will be flash frozen at 24 hr post-mating (the time at which we have seen maximal differential transcript usage response). Virgin females will be collected as controls. RNA will be extracted from female heads, and standard polyA-selected RNA-seq libraries will be produced, being sure to do paired-end to maximize informativeness of alternative splice forms. We will use Salmon software for computational inference of transcript quantification (77). Since we're seeing alternative promoter use after mating, a method that focuses on transcription start sites will also be used (e.g. MAPCAP; (78)). We will quantify the level of alternative exon use in each combination; if there is variation in alternative exon use, we will focus our analysis on *foraging* and *Desat1*.

These data will lend themselves well to analysis with well-understood general linear models, asking to quantify for each transcript of each gene the significance of among-line variance in male effects, female effects and interaction terms. Because we have de novo assemblies of whole-genomes of the DSPR lines, we will be able to determine whether there are effects of SNPs within promoter regions of alternatively used promoters. In addition to this approach, the models that are specifically geared for “pairwise contests,” described in Aim 3, will also be applied. Here we not only ask about the male and female effects, but we can also ask if knowledge of a subset of the pairings allows prediction of other pairings. This is based on a model that assumes that male effects and female effects fall on some linear ranking. Rejection of fit of this class of model indicates a specific sort of pairwise interaction. Aggregated across many genes, and these results could reveal important patterns through combinations of genes that consistently fit poorly. See Aim 3 for more details.

Once we know from the DSRP studies whether male, female, or both are major drivers of the alterative exon use, we will map the factors that cause the differences using a carefully chosen subset of the DSPR recombinant inbred lines, scoring both transcript use and sperm competition. Candidate genes in the QTL intervals that show a significant correlation will then be knocked out, using CRISPR/Cas9 (27, 66, 79) or, if knockouts are inviable or subviable, knocked down in the brain as described in Aim 1 but using brain-specific drivers in addition to the pan-neuronal elav-GAL4. We will then use qRT-PCR on crosses with knockouts/knockdowns and controls to test whether the candidate gene affects exon usage in the female’s brain after mating. If it does, then this gene is part of the pathway through which Sex Peptide exerts its effects. We will determine whether these genes influence sperm competition outcomes and post-mating behaviors, and will place them within a pathway from Sex Peptide to splicing/promoter choice.

The data from this sub-Aim will also let us determine if there are correlates between alternative exon use and other reproduction attributes. The primary functional classes of genes seen in the post-mating transcriptional response are immunity and metabolism, and a number of the genes with altered transcript use also fall in these biological function groups. It will also be essential to follow up with targeted assays of key genes in materials derived from judiciously chosen crosses. We will also pursue analysis of gene networks that appear to show coordinated changes in response to mating. In the end, this experiment will provide rigorous quantification of the magnitudes of male, female and interaction effects on the control of differential transcript use post-mating, and the degree to which those responses are correlated with whole-organism reproductive phenotypes.

Aim 2b: How do alterations in specific genes’ mating-induced changes in alternative exon use in the female impact the male’s fertilization success?

Analysis of our post-mating transcriptional time course revealed a few particularly responsive genes that warrant further analysis and follow-up. When we quantified exon-specific expression of the post-mating time-course data mentioned at the beginning of this aim, we saw many instances of what appeared to be either alternative splicing or alternative promoter usage. Production or expression of different proteins due to alternative splicing and/or alternative promoter use is important in generating distinct biological outcomes (31-33). For example, alternative splicing of the *dsx* transcript results in the male-specifying vs female-specifying forms of this major sex-regulator (80), causing development of males instead of females, or vice versa. Dsx was recently shown to also play a role in post-mating behaviors: depletion of Dsx protein in specific neurons of virgin females drives behaviors characteristic of mated females (81).

One exciting discovery is the mating-induced alternative exon use of the *foraging* gene. Promoter 4 has been studied in detail in adults, where its products were found to influence foraging behavior. Specifically, a reduction in the abundance of transcripts from promoter 4 makes foraging behavior more “rover-like” (animals have a longer path-length when searching for food); indeed “rover” strains of Drosophila have lower expression of promoter 4 in the brain. (82)). We hypothesize that *foraging* has something to do with female food intake after mating, or perhaps memory/sleep, both of which may have indirect effects on the female’s reproduction, including indirectly on her selectivity of sperm.

We will test whether the post-mating alternative promoter use that we discovered is biologically meaningful and whether it is affected by male x female genotype interactions. We will verify and quantify the alternative promoter use by qRT-PCR with internal controls (RPL32) at our post-mating time points. Then, using *for* mutants and transgenic *for* cDNA expression lines that were previously used to determine the function of the *foraging* isoforms (1), we will test whether expression of the promoter 4 isoforms of *for* affect the female’s post-mating responses (e.g. feeding, sleep, mate-searching) by assays that are standard for us (83, 84), and whether they modulate the outcome of sperm competition (13, 34, 53). Dr. Marla Sokolowski has made all of the needed fly lines available to us for these experiments (please see letter from Dr. Sokolowski).

We will obtain the published PR5-Gal4 line from Bousquet et al. (76) and use it to knockdown *Desat1*. Standard phenotypic assays for mate preference/choice and receptivity (8, 83) and sperm competition outcomes (13, 34, 53) will allow direct confirmation of the role of *Desat1* in those specific phenotypes.

Potential challenges and alternative approaches

The thorough quantification of genetic variation in post-mating alternative exon use (including alternative promoter use and alternative splicing) faces no technical hurdles, and is easily replicated to assure quality data. While we do not yet know whether or not the splicing and promoter-use changes that we identify are a direct consequence of Sex Peptide action, our identification of genes that lead to them – directly or indirectly – in our the GWAS experiments will be useful and novel in connecting mating to alternate exon use.
The challenge in this aim is in associating those transcriptome responses to differences in reproductive success. This will have to be correlative at the start, testing by statistical association among fly lines the relationship between transcriptome attributes and sperm competition and other reproductive phenotypes. But once we have a collection of these associations, the validation step of assessing the phenotypic consequences of targeted alterations in the transcriptome will follow standard approaches with which we have been successful in the future. Two particularly compelling examples will be tested in sub-Aim 2b. Our studies of *foraging* will synergize with what is known about this gene's role in learning and searching/risk-taking.

Aim 3: Apply Bradley-Terry models to assess the nature of pairwise interactions in mating.

Sperm competition begins as a pairwise competition – a female mates with two males in succession, and then she has the opportunity to assess the males' qualities and regulate differential use of their sperm. In fact, a great many aspects of reproduction involve this sort of A/B decision, and our assessment of these phenomena typically involve experiments that have a large series of pairwise contests. When presented with a large number of pairwise contests among many distinct players, it is possible that the past record of successes provides an ability to predict the outcome of an as-yet-untested pairwise contest. However, some contests may fail to adhere to such a simple model. Careful modeling of the data from pairwise contests could identify interactions among specific genes and specific reproductive attributes more precisely than typical linear models. The following three sub-aims develop such an approach for evaluating the "transitivity" of sperm competitive interactions and their regulators.

Aim 3a Apply the Bradley-Terry model to pairwise sperm competition tests with multiple distinct male genotypes.

The Bradley and Terry model (30) is one of the many probability models that estimate player ratings from a series of pairwise contests and fit a model that also allows estimation of the probability that one player (or team) beats another (85, 86). Player ratings are an estimate of competitiveness of each individual and are calculated based on outcomes of past pairwise contests. Another ranking method, the Élő ranking system (87), was developed later and independently to provide a means for ranking the performance of chess players by assigning scores to players. Mathematically these two models are actually equivalent, and henceforth we will refer only to the Bradley-Terry model ("B-T").

The B-T model can be used to test whether there is a simple ranking of players that display stochastic transitivity and to make probabilistic predictions of outcomes of matches that have not yet occurred. So if A has beaten B two-thirds of the time, and B has beaten C two-thirds of the time, then not only do we expect A to beat C (simple transitivity), but in fact the model predicts that A will beat C with probability 0.8 (simply solve for C/(A+C) when A/(A+B) = 2/3, and B/(B+C) = 2/3). In more general terms, if the quality or competitiveness of player *i* is α_i, and that of player *j* is $\alpha\ \alpha_j$, then the probability that *i* beats *j* is $\alpha_i / (\alpha_i + \alpha_j)$. The breakthrough of the Bradley and Terry paper (30) was to show that there is a closed-form maximum likelihood solution, given a large series of contest results (win-loss records), one can estimate the best fitting values of α_i for all players. Also of great interest is that this solution provides a goodness-of-fit test. A rejection of the model indicates that the simple rank-ordering is not preserved and is not predictive of outcomes, i.e. there are specific pairwise interactions that result in deviations from the model, implying that the trait we are looking at is not the simple sole predictor of sperm competition outcome.

The question of transitivity of behaviors arises frequently in population biology (88), and while it seems like a rational approach, there are situations both theoretical and empirical that produce nontransitive behaviors (89). Application of the B-T model is less common, but it too has appeared in the population biology literature. Shev

et al. (90) applied the B-T model to obtain fits to the dominance hierarchy among rhesus macaques engaging in pairwise contests. Muller et al. (91) obtained estimates of host preference of saproxylic beetles with the B-T model, and they assessed the effect of covariates on preferences with a method of recursive partitioning.

We will apply the B-T model and develop extensions to obtain estimates of intrinsic sperm competitive abilities for each genotype and assess goodness-of-fit of the model. This modeling framework has a natural application for estimating parameters relevant to sperm competitiveness when the experimental data are in the form of pairwise contests. Our double-mating assays are exactly this, and most of the other assays, including mating plug ejection timing and alternative exon use can be expressed in terms that allow application of the model. Parameters of the model assign a competitiveness value to each genotype and these in turn can be given confidence limits. These estimates are readily interpreted as relative sperm competitive abilities, and the differences in these can easily be used to infer allele frequency dynamics in a population (assuming sperm competition is the only component of fitness that varies).

As a proof-of-principle, we apply the B-T model to mate-choice data from Arbuthnott et al. (92) where 11 DGRP line females were competed in the 55 possible pairs for matings. Fig. 6 shows that the B-T model estimates stabilize over successive bouts of mating, after about 200 trials. The model also rejects the null model of no mating preference differences at $P < 6.29 \times 10^{-36}$, and shows an excellent fit to the transitive model. In this sub-aim we will assess approaches that allow accommodation for covariates. The simplest approach is to remove the effect of the covariates up front with a linear model, and run the B-T on the residuals. We will simulate the process of the pairwise contests with inclusion of significant covariates to assess performance of this approach. We will also generate data with known deviation from transitivity to quantify the power to detect model deviation.

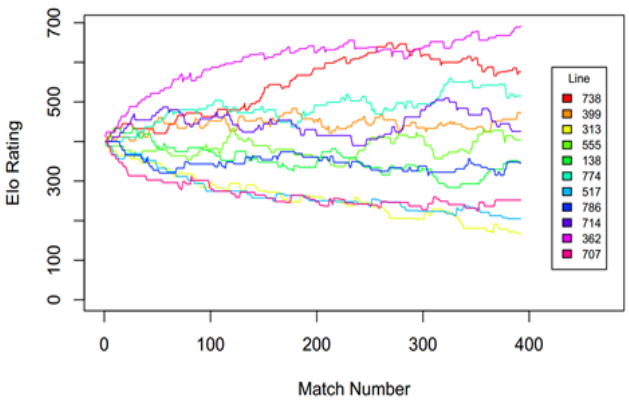


Fig. 6 Fits of the Bradley-Terry/Elo model to male-choice mating experiment data, showing that rank orders of mate preferences stabilize reasonably quickly. (Data from Arbuthnott et al. (2017)).

After the B-T model was published, a large number of models made extensions to the approach, seeking to account for non-independence (reviewed in (93)). The extensions allow for covariates that may also impact the experiment outcome, such that ignoring the covariate could distort the results reported by the simpler B-T model. In the experiments we do, this consideration of covariates is important, because it allows for accommodation of such technical experimental factors as batch effects, replicate day (accounting for micro-climate effects, food-batch effects, etc.), as well as potentially important biological factors such as age, previous mating experience of the male, and male body size.

Aim 3b Apply generalized extensions of pairwise contest models to a variety of phenotypes related to mating and gametic success

The generalized B-T model will be fitted to a wide variety of experiments, including those that we will perform in Aims 1 and 2, as well as others that have already been performed in our labs, to assess sperm competitive success in its many components. In addition to sperm competition outcomes (paternity), we will apply the models to timing of mating plug ejection, differential gene expression and alternative exon use in females after mating (94). An additional set of covariates that we will also include is allelic state of the major seminal proteins that are known to affect sperm storage or competition outcomes, including Sex Peptide, Acp36DE, Acp29AB, Acp62F (59-61, 95, 96); their allelic state is known from publicly available DGRP sequences. Results will allow us to assess the significance of differences among lines in each of these attributes. In the simplest case, all of these factors would fit the B-T models acceptably, and simple rank orderings of genotypes would be consistent and informative for predicting outcomes of the pairwise contests. We have previously reported instances of strong interactions, such as when the rank ordering of males in their sperm competitive abilities depended on the female genotype as well (97).

Aim 3c Use the results of pairwise contest statistics to quantify male x female interactions and develop an understanding of the role of those interactions in mating success.

In this sub-Aim we will integrate the results from sub-Aim 3b and draw biological inferences from the models. For example, we will have predictions of each of the DSPR line's male competitiveness with respect to overall sperm competitive ability (the grid crosses in Aims 1 and 2), as well as their ability to regulate the timing of

mating plug ejection by the female. To the extent that sperm competitiveness is determined by mating plug ejection timing, these should be highly correlated, and we can test this directly. Similarly, we will be able to assess whether levels of alternative exon use in the brain are informative for improving predictions of sperm competition outcomes. In cases where there are breakdowns of transitivity, we will be able to identify which lines are the most responsible, and identify correlates that explain the breakdown.

Potential challenges and alternative approaches

The B-T model begins with fairly simple assumptions about the rank-ordering and stability of competitiveness of each “player,” but one still has to be careful about conclusions drawn from the fits. It requires additional validations beyond seeing a good fit of a model to the data before one is convinced that all of the assumptions of the model are true. But the B-T model provides a rigorous starting point for understanding and even predicting outcomes of pairwise contests. Whenever we encounter situations that seem counterintuitive, we will be able to resort to extensive simulations, as will be used in sub-Aim 3b, to explore the behavior of the model. In many circumstances one runs an experiment at a sample size that is expected to have adequate power, but in the case of pairwise contests like sperm competition assays, the B-T model’s ability to predict outcomes of pairwise mating experiments provides a strong intuitive confirmation that the statistical power is adequate.

Concluding remarks

Through the auspices of this grant, we have identified genetic variation in, and interactions between, males and females that controls the relative fertility success of two competing males when mating with different females. We have shown the active contribution of the female, through her nervous system, and identified specific neuron-types, and genes, needed for this. We have shown that male and female contributions to the regulation of release of sperm from storage and to changes in female behavior after mating can advantage a male’s relative reproductive success. In the present funding period two of the three Aims that we propose continue this analysis of dissecting the male- and female- contributions to sperm competition and the interactions between them. First, we will determine the genetic and neural basis for the female’s ejection of the mating plug, which controls the number of a male’s sperm available to fertilize her eggs, and the male’s effect on these. Second, we will follow up on our recent exciting finding that mating (and a particular seminal protein) controls alternative promoter- and splice-site use in the female’s brain. We will determine whether this is driven by the male’s or female’s genotype, or interactions between them, and identify the underlying genes that participate in this phenomenon. We will then test the roles of these genes or their specific targets in behaviors that affect the outcome of sperm competition. Finally, we will analyze these results using a predictive model that is uniquely suited to the “pairwise contest” aspect of the experiments, providing a better picture of the role of male x female interactions in these important attributes of reproductive biology.

Rigor, reproducibility, and transparency

Drosophila stocks are validated prior to each experiment by using visible markers, PCR, or RT-PCR for knockdowns. Experiments are carried out with all necessary controls, and in multiple biological replicates, and with sample sizes pre-determined to provide the needed power. We use standardized experimental setups (food, temperature, light-regimens) for all experiments. Where possible, we randomize flies for assays so that results can be obtained double-blind. Data analysis and appropriate statistical tests are performed in R. All data and all scripts are made publicly available upon publication.

Time Line

Specific Aim 1 will begin with completion of the GWAS for mating plug ejection timing in year 1. The 6x6 grid cross of DSPR lines for assessing determinants of mating plug ejection timing will be done in year 2. We will launch the validation and knockdown experiments in years 2-4. Subsequent follow-up studies will round out years 4 and 5. Specific Aim 2 will begin in year 1 with the studies of post-mating effects on alternative splicing and alternative promoter use. Beginning in year 2, and continuing through years 3 and 4, we will screen through the DGRP lines and perform the 6x6 DSPR grid cross for differential post-mating transcript/exon use. Specific Aim 3 will start in year 1 with the model development to handle covariates, and in years 2-4 we will run the B-T models and their derivatives on all of our suitable sperm competition, mating plug timing, post-mating gene expression data. In years 4 and 5 we will integrate the results of these model fits as described in Aim 3. In year 5 we will wrap up this analysis and the writing.

PROGRESS REPORT PUBLICATIONS – HD059060 – 2016-2021

Ågren JA, Munasinghe M, Clark AG. Sexual conflict through mother's curse and father's curse. Theor Popul Biol. 2019 Oct;129:9-17. doi: 10.1016/j.tpb.2018.12.007. Epub 2019 May 2. PMID: 31054851.

Ahmed-Braimah YH, Unckless RL, Clark AG. Evolutionary Dynamics of Male Reproductive Genes in the *Drosophila virilis* Subgroup. G3 (Bethesda). 2017 Sep 7;7(9):3145-3155. doi: 10.1534/g3.117.1136. PMID: 28739599; PMCID: PMC5592939.

Ahmed-Braimah YH, Wolfner MF, Clark AG. Differences in Postmating Transcriptional Responses between Conspecific and Heterospecific Matings in Drosophila. Mol Biol Evol. 2021 Mar 9;38(3):986-999. doi: 10.1093/molbev/msaa264. PMID: 33035303; PMCID: PMC7947788.

Allen SE, Chen D, Misra S, Wolfner MF. Seminal Metalloprotease-1. Handbook of Proteolytic Enzymes, 4th edition. 2021 Vol. 1 Metalloproteases. N. Rawlings, editor. Elsevier. In press.

Anholt RRH, O'Grady P, Wolfner MF, Harbison ST. Evolution of Reproductive Behavior. Genetics 2020 Jan;214(1):49-73. doi: 10.1534/genetics.119.302263. PMID: 31907301; PMCID: PMC6944409.

Billeter J-C, Wolfner MF. Chemical cues that guide female reproduction in *Drosophila melanogaster*. J Chem Ecol. 2018 Sep;44(9):750-769. doi: 10.1007/s10886-018-0947-z. PMID: 29557077; PMCID: PMC6085157.

Chapman T, Wolfner MF. Reproductive behaviour: Make love, then war. Nat Ecol Evol. 2017 May 15;1(6):174. doi: 10.1038/s41559-017-0174. PMID: 28812638; PMCID: PMC5745578.

Chen DS, Delbare SYN, White SL, Sitnik J, Chatterjee M, DoBell E, Weiss O, Clark AG, Wolfner MF. Female Genetic Contributions to Sperm Competition in *Drosophila melanogaster*. Genetics. 2019 Jul;212(3):789-800. doi: 10.1534/genetics.119.302284. Epub 2019 May 17. PMID: 31101677; PMCID: PMC6614900.

Chow CY, Kelsey K, Wolfner MF, Clark AG. Candidate genetic modifiers of retinitis pigmentosa identified by exploiting natural variation in *Drosophila*. Hum Mol Genet. 2016 Feb 15;25(4):651-9. doi: 10.1093/hmg/ddv502. Epub 2015 Dec 11. PMID: 26662796; PMCID: PMC4743685.

Delbare SYN, Chow CY, Wolfner MF, Clark AG. Roles of Female and Male Genotype in Post-Mating Responses in Drosophila melanogaster. J Hered. 2017 Oct 30;108(7):740-753. doi: 10.1093/jhered/esx081. PMID: 29036644; PMCID: PMC5892397.

Delbare SYN, Ahmed-Braimah YH, Wolfner MF, Clark AG. Interactions between the microbiome and mating influence the female's transcriptional profile in Drosophila melanogaster. Sci Rep. 2020 Oct 23;10(1):18168. doi: 10.1038/s41598-020-75156-9. PMID: 33097776; PMCID: PMC7584617.

Giardina TJ, Clark AG, Fiumera AC. Estimating mating rates in wild *Drosophila melanogaster* females by decay rates of male reproductive proteins in their reproductive tracts. Mol Ecol Resour. 2017 Nov;17(6):1202-1209. doi: 10.1111/1755-0998.12661. Epub 2017 Mar 24. PMID: 28213940.

Hafezi Y, Sruba SR, Tarrash SR, Wolfner MF, Clark AG. Dissecting Fertility Functions of *Drosophila Y* Chromosome Genes with CRISPR. Genetics. 2020 Apr;214(4):977-990. doi: 10.1534/genetics.120.302672. Epub 2020 Feb 25. PMID: 32098759; PMCID: PMC7153933.

Hoke K, Adkins-Regan E, Bass AH, McCune AR, Wolfner MF. Co-opting Evo-Devo concepts for new insights into behavioural and neural diversity. J Exp Biol. 2019 Apr 15;222(Pt 8):jeb190058. doi: 10.1242/jeb.190058. PMID: 30988051; PMCID: PMC6503947

Immarigeon C, Frei Y, Delbare SYN, Gligorov D, Machado Almeida P, Grey J, Fabbro L, Nagoshi E, Billeter JC, Wolfner MF, Karch F, Maeda RK. Identification of a micropeptide and multiple secondary cell genes that

modulate *Drosophila* male reproductive success. Proc Natl Acad Sci U S A. 2021 Apr 13;118(15):e2001897118. doi: 10.1073/pnas.2001897118. PMID: 33876742; PMCID: PMC8053986.

Maeda RK, Sitnik JL, Frei Y, Prince E, Gligorov D, Wolfner MF, Karch F. The lncRNA male-specific abdominal plays a critical role in Drosophila accessory gland development and male fertility. PLoS Genet. 2018 Jul 16;14(7):e1007519. doi: 10.1371/journal.pgen.1007519. PMID: 30011265; PMCID: PMC6067764.

Meyer H, Buhr A, Callaerts P, Scheimann R, Wolfner MF, Marygold SJ. Identification and bioinformatic analysis of neprilysin and neprilysin-like metalloendopeptidases in *Drosophila melanogaster*. MicroPubl Biol. 2021 Jun 23;2021:10.17912/micropub.biology.000410. doi: 10.17912/micropub.biology.000410. PMID: 34189422; PMCID: PMC8223033.

Orr TJ, Burns M, Hawkes K, Holekamp KE, Hook KA, Josefson CC, Kimmitt AA, Lewis AK, Lipshutz SE, Lynch KS, Sirot LK, Stadtmauer DJ, Staub NL, Wolfner MF, Hayssen V. It Takes Two to Tango: Including a Female Perspective in Reproductive Biology. Integr Comp Biol. 2020 Sep 1;60(3):796-813. doi: 10.1093/icb/icaa084. PMID: 32702091.

Sirot LK, Wolfner MF. "Call and Response": A Case of Behavioral-Molecular Copulatory Dialogue? Bioessays. 2020 Nov;42(11):e2000248. doi: 10.1002/bies.202000248. Epub 2020 Oct 5. PMID: 33017061

Sitnik JL, Gligorov D, Maeda RK, Karch F, Wolfner MF. The Female Post-Mating Response Requires Genes Expressed in the Secondary Cells of the Male Accessory Gland in Drosophila melanogaster. Genetics. 2016 Mar;202(3):1029-41. doi: 10.1534/genetics.115.181644. Epub 2016 Jan 8. PMID: 26746709; PMCID: PMC4788108.

White MA, Chen DS, Wolfner MF. She's got nerve: roles of octopamine in insect female reproduction. J Neurogenet. 2021 Apr 28:1-22. doi: 10.1080/01677063.2020.1868457. Epub ahead of print. PMID: 33909537.

Wigby S, Brown NC, Allen SE, Misra S, Sitnik JL, Sepil I, Clark AG, Wolfner MF. The *Drosophila* seminal proteome and its role in postcopulatory sexual selection. Philos Trans R Soc Lond B Biol Sci. 2020 Dec 7;375(1813):20200072. doi: 10.1098/rstb.2020.0072. Epub 2020 Oct 19. PMID: 33070726; PMCID: PMC7661438.

PHS Human Subjects and Clinical Trials Information

OMB Number: 0925-0001

Expiration Date: 02/28/2023

Use of Human Specimens and/or Data

Does any of the proposed research in the application involve human specimens and/or data *	❍ Yes ● No
Provide an explanation for any use of human specimens and/or data not considered to be human subjects research.	
Are Human Subjects Involved	❍ Yes ● No
Is the Project Exempt from Federal regulations?	❍ Yes ❍ No
Exemption Number	❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8
Other Requested Information	

MULTIPLE PI LEADERSHIP PLAN

Both P.I.s are members of the Department of Molecular Biology and Genetics at Cornell, and both are also members of the Graduate Fields of Genetics, Genomics & Development, and of Ecology & Evolutionary Biology. Both are also members of the Cornell Center for Comparative and Population Genomics. We are located in the same building, two floors apart. Our labs have a record of informal and formal collaboration for over 25 years – beginning when we were at different institutions (Penn State, AC; Cornell, MW). As an example of informal (non-co-authored) publications, Clark *et al.* (1995) used Wolfner-lab's Acp gene sequences, prior to their (1997) publication, and Clark did the calculation for statistical significance of the Acp genes' distribution for Wolfner *et al.* (1997). Formal collaborations leading to publications include: Swanson *et al.* (2001a), McGraw *et al.* (2004, 2007, 2008, 2009), Mueller *et al.* (2005), and Chow *et al*. (2010, 2013a,b, 2015a,b, 2016), Delbare *et al.* (2017, 2020), Chen *et al.* (2019), Wigby *et al.* (2020), Hafezi *et al.* (2020), Ahmed-Braimah *et al.* (2021). We have also co-taught a course on the genomics of non-model organisms. There are many ongoing informal interactions between the two PIs, between each of us with the other's lab members, and among members of the two labs. We have co-mentored graduate students formally (Brown, Chen, Delbare, White) and informally (McGraw, Mueller), and have co-mentored postdoctoral fellows formally (Chow) and informally (Hafezi). We have been joint PIs on this NIH grant for 12 years, and also on internal grants (Cornell Center for Vertebrate Genomics).

For this proposal: the "Global Diversity" strains that were derived from natural populations were established in and are maintained by the Clark lab. AGC maintains the genome sequence database on these lines. Transgenic and mutant strains are maintained in the Wolfner lab, or obtained from the Bloomington or Vienna Drosophila Stock Centers; additional strains will be generated and maintained in the Wolfner lab. The Wolfner lab has had success generating mutations with CRISPR technologies, and MFW will provide oversight expertise on this aspect of the proposed studies as well as and on molecular/microscopy, and reproductive assays, and aspects of Aim 1 that center on neuronal function and manipulation. AGC will supervise the GWAS analysis of Aim 1. Both labs have extensive experience in all phases of RNA-seq analysis (as do many lab members now), and both will supervise the alternate transcript component of Aim 2. AGC has more experience in modeling and has worked with the Bradley-Terry model, and he will supervise Aim 3. Both PIs are experienced with Drosophila sperm competition assays and will jointly supervise those. Similarly, both will be actively involved in the planning, writing, and revising of manuscripts, working with the students and postdocs that are the first-authors.

All project personnel will go back and forth between the labs regularly, and attend lab meetings of both labs (so that they receive training in areas of expertise of both groups). Project personnel have benches in both labs, although they may spend extended time in one lab or the other depending on what is appropriate for a given analysis. Formal weekly meetings for the project group will continue to be held throughout the duration of the study to discuss results, plans, paper-preparation, etc.

BIBLIOGRAPHY

1. Allen AM, Anreiter I, Neville MC, Sokolowski MB. Feeding-Related Traits Are Affected by Dosage of the foraging Gene in Drosophila melanogaster. Genetics. 2017;205(2):761-73. Epub 2016/12/23. doi: 10.1534/genetics.116.197939. PubMed PMID: 28007892; PMCID: PMC5289850.
2. Delbare SYN, Venatramen S, Wells MT, Wolfner, MF, Basu S, Clark AG. Mating and Sex Peptide influence RNA abundance and exon use in heads of Drosophila melanogaster females. in prep.
3. Wigby S, Brown NC, Allen SE, Misra S, Sitnik JL, Sepil I, Clark AG, Wolfner MF. The Drosophila seminal proteome and its role in postcopulatory sexual selection. Philos Trans R Soc Lond B Biol Sci. 2020;375(1813):20200072. Epub 2020/10/20. doi: 10.1098/rstb.2020.0072. PubMed PMID: 33070726; PMCID: PMC7661438.
4. Parker GA. Conceptual developments in sperm competition: a very brief synopsis. Philos Trans R Soc Lond B Biol Sci. 2020;375(1813):20200061. Epub 2020/10/20. doi: 10.1098/rstb.2020.0061. PubMed PMID: 33070727; PMCID: PMC7661437.
5. Aalberts M, Stout TA, Stoorvogel W. Prostasomes: extracellular vesicles from the prostate. Reproduction. 2014;147(1):R1-14. doi: 10.1530/REP-13-0358. PubMed PMID: 24149515.
6. Corrigan L, Redhai S, Leiblich A, Fan SJ, Perera SM, Patel R, Gandy C, Wainwright SM, Morris JF, Hamdy F, Goberdhan DC, Wilson C. BMP-regulated exosomes from Drosophila male reproductive glands reprogram female behavior. The Journal of cell biology. 2014;206(5):671-88. doi: 10.1083/jcb.201401072. PubMed PMID: 25154396; PMCID: 4151142.
7. Mackay TF, Richards S, Stone EA, Barbadilla A, Ayroles JF, Zhu D, Casillas S, Han Y, Magwire MM, Cridland JM, Richardson MF, Anholt RR, Barron M, Bess C, Blankenburg KP, Carbone MA, Castellano D, Chaboub L, Duncan L, Harris Z, Javaid M, Jayaseelan JC, Jhangiani SN, Jordan KW, Lara F, Lawrence F, Lee SL, Librado P, Linheiro RS, Lyman RF, Mackey AJ, Munidasa M, Muzny DM, Nazareth L, Newsham I, Perales L, Pu LL, Qu C, Ramia M, Reid JG, Rollmann SM, Rozas J, Saada N, Turlapati L, Worley KC, Wu YQ, Yamamoto A, Zhu Y, Bergman CM, Thornton KR, Mittelman D, Gibbs RA. The Drosophila melanogaster Genetic Reference Panel. Nature. 2012;482(7384):173-8. doi: 10.1038/nature10811. PubMed PMID: 22318601; PMCID: 3683990.
8. Kohlmeier P, Zhang Y, Gorter JA, Su CY, Billeter JC. Mating increases Drosophila melanogaster females' choosiness by reducing olfactory sensitivity to a male pheromone. Nat Ecol Evol. 2021. Epub 2021/06/23. doi: 10.1038/s41559-021-01482-4. PubMed PMID: 34155384.
9. Wang K, Wang F, Forknall N, Yang T, Patrick C, Parekh R, Dickson BJ. Neural circuit mechanisms of sexual receptivity in Drosophila females. Nature. 2021;589(7843):577-81. Epub 2020/11/27. doi: 10.1038/s41586-020-2972-7. PubMed PMID: 33239786.
10. Delbare SYN, Chow CY, Wolfner MF, Clark AG. Roles of Female and Male Genotype in Post-Mating Responses in Drosophila melanogaster. J Hered. 2017;108(7):740-53. Epub 2017/10/17. doi: 10.1093/jhered/esx081. PubMed PMID: 29036644; PMCID: PMC5892397.
11. Delbare SYN, Wolfner MF, Clark AG. Mating-responsive RNAs in the reproductive tract of Drosophila melanogaster. in prep.
12. Delbare SYN, Ahmed-Braimah YH, Wolfner MF, Clark AG. Interactions between the microbiome and mating influence the female's transcriptional profile in Drosophila melanogaster. Sci Rep. 2020;10(1):18168. Epub 2020/10/25. doi: 10.1038/s41598-020-75156-9. PubMed PMID: 33097776; PMCID: PMC7584617.
13. Chen DS, Delbare SYN, White SL, Sitnik J, Chatterjee M, DoBell E, Weiss O, Clark AG, Wolfner MF. Female Genetic Contributions to Sperm Competition in Drosophila melanogaster. Genetics. 2019;212(3):789-800. Epub 2019/05/19. doi: 10.1534/genetics.119.302284. PubMed PMID: 31101677; PMCID: PMC6614900.
14. Seeholzer LF, Seppo M, Stern DL, Ruta V. Evolution of a central neural circuit underlies Drosophila mate preferences. Nature. 2018;559(7715):564-9. Epub 2018/07/12. doi: 10.1038/s41586-018-0322-9. PubMed PMID: 29995860; PMCID: PMC6276375.

15. Ahmed-Braimah YH, Wolfner MF, Clark AG. Differences in Postmating Transcriptional Responses between Conspecific and Heterospecific Matings in Drosophila. Molecular biology and evolution. 2021;38(3):986-99. Epub 2020/10/10. doi: 10.1093/molbev/msaa264. PubMed PMID: 33035303; PMCID: PMC7947788.
16. Immarigeon C, Frei Y, Delbare SYN, Gligorov D, Machado Almeida P, Grey J, Fabbro L, Nagoshi E, Billeter JC, Wolfner MF, Karch F, Maeda RK. Identification of a micropeptide and multiple secondary cell genes that modulate Drosophila male reproductive success. Proceedings of the National Academy of Sciences of the United States of America. 2021;118(15). Epub 2021/04/21. doi: 10.1073/pnas.2001897118. PubMed PMID: 33876742; PMCID: PMC8053986.
17. Maeda RK, Sitnik JL, Frei Y, Prince E, Gligorov D, Wolfner MF, Karch F. The lncRNA male-specific abdominal plays a critical role in Drosophila accessory gland development and male fertility. PLoS genetics. 2018;14(7):e1007519. Epub 2018/07/17. doi: 10.1371/journal.pgen.1007519. PubMed PMID: 30011265; PMCID: PMC6067764.
18. Sitnik JL, Gligorov D, Maeda RK, Karch F, Wolfner MF. The Female Post-Mating Response Requires Genes Expressed in the Secondary Cells of the Male Accessory Gland in Drosophila melanogaster. Genetics. 2016;202(3):1029-41. doi: 10.1534/genetics.115.181644. PubMed PMID: 26746709; PMCID: 4788108.
19. Chapman T, Wolfner MF. Reproductive behaviour: Make love, then war. Nat Ecol Evol. 2017;1(6):174. Epub 2017/08/16. doi: 10.1038/s41559-017-0174. PubMed PMID: 28812638; PMCID: PMC5745578.
20. Billeter JC, Wolfner MF. Chemical Cues that Guide Female Reproduction in Drosophila melanogaster. J Chem Ecol. 2018;44(9):750-69. Epub 2018/03/21. doi: 10.1007/s10886-018-0947-z. PubMed PMID: 29557077; PMCID: PMC6085157.
21. Hoke KL, Adkins-Regan E, Bass AH, McCune AR, Wolfner MF. Co-opting evo-devo concepts for new insights into mechanisms of behavioural diversity. J Exp Biol. 2019;222(Pt 8). Epub 2019/04/17. doi: 10.1242/jeb.190058. PubMed PMID: 30988051; PMCID: PMC6503947.
22. Anholt RRH, O'Grady P, Wolfner MF, Harbison ST. Evolution of Reproductive Behavior. Genetics. 2020;214(1):49-73. Epub 2020/01/08. doi: 10.1534/genetics.119.302263. PubMed PMID: 31907301; PMCID: PMC6944409.
23. Sirot LK, Wolfner MF. "Call and Response": A Case of Behavioral-Molecular Copulatory Dialogue? Bioessays. 2020;42(11):e2000248. Epub 2020/10/06. doi: 10.1002/bies.202000248. PubMed PMID: 33017061.
24. Meyer H, Buhr A, Callaerts P, Schiemann R, Wolfner MF, Marygold SJ. Identification and bioinformatic analysis of neprilysin and neprilysin-like metalloendopeptidases in Drosophila melanogaster. MicroPubl Biol. 2021;2021. Epub 2021/07/01. doi: 10.17912/micropub.biology.000410. PubMed PMID: 34189422; PMCID: PMC8223033.
25. Allen SE, Chen DS, Misra S, Wolfner M. Seminal Metalloprotease-1. In: Rawlings N, editor. Handbook of Proteolytic Enzymes. 4th ed: Elsevier; 2021. p. in press.
26. White MA, Chen DS, Wolfner MF. She's got nerve: roles of octopamine in insect female reproduction. J Neurogenet. 2021:1-22. Epub 2021/04/29. doi: 10.1080/01677063.2020.1868457. PubMed PMID: 33909537.
27. Hafezi Y, Sruba SR, Tarrash SR, Wolfner MF, Clark AG. Dissecting Fertility Functions of Drosophila Y Chromosome Genes with CRISPR. Genetics. 2020;214(4):977-90. Epub 2020/02/27. doi: 10.1534/genetics.120.302672. PubMed PMID: 32098759; PMCID: PMC7153933.
28. Chow CY, Kelsey KJ, Wolfner MF, Clark AG. Candidate genetic modifiers of retinitis pigmentosa identified by exploiting natural variation in Drosophila. Human molecular genetics. 2016;25(4):651-9. doi: 10.1093/hmg/ddv502. PubMed PMID: 26662796; PMCID: 4743685.
29. Holman A, Munasinghe M, Clark AG. Natural Polymorphism in Second-site Modifiers of the Drosophila C9orf72 Model for Amyotrophic Lateral Sclerosis. . Genetics Neuroscience.in revision.
30. Bradley RA, Terry ME. Rank analysis of incomplete block designs: I. The method of paired comparisons. . Biometrika. 1952;39 ((3/4)):324.

31. Angarola BL, Anczukow O. Splicing alterations in healthy aging and disease. Wiley Interdiscip Rev RNA. 2021;12(4):e1643. Epub 2021/02/11. doi: 10.1002/wrna.1643. PubMed PMID: 33565261; PMCID: PMC8195850.
32. Kolathur KK. Role of promoters in regulating alternative splicing. Gene. 2021;782:145523. Epub 2021/03/06. doi: 10.1016/j.gene.2021.145523. PubMed PMID: 33667606.
33. Venables JP, Tazi J, Juge F. Regulated functional alternative splicing in Drosophila. Nucleic acids research. 2012;40(1):1-10. Epub 2011/09/13. doi: 10.1093/nar/gkr648. PubMed PMID: 21908400; PMCID: PMC3245913.
34. Chow CY, Wolfner MF, Clark AG. Large neurological component to genetic differences underlying biased sperm use in Drosophila. Genetics. 2013;193(1):177-85. doi: 10.1534/genetics.112.146357. PubMed PMID: 23105014; PMCID: 3527244.
35. Chen DS, Clark AG, Wolfner MF. Octopamine is necessary for sperm preference in doubly-mated Drosophila females. in prep.
36. Rubinstein CD, Wolfner MF. Drosophila seminal protein ovulin mediates ovulation through female octopamine neuronal signaling. Proceedings of the National Academy of Sciences of the United States of America. 2013;110(43):17420-5. doi: 10.1073/pnas.1220018110. PubMed PMID: 24101486; PMCID: 3808635.
37. Avila FW, Bloch Qazi MC, Rubinstein CD, Wolfner MF. A requirement for the neuromodulators octopamine and tyramine in Drosophila melanogaster female sperm storage. Proceedings of the National Academy of Sciences of the United States of America. 2012;109(12):4562-7. Epub 2012/03/07. doi: 10.1073/pnas.1117689109. PubMed PMID: 22393023; PMCID: 3311333.
38. Chen DS, Clark AG, Wolfner MF. Octopaminergic signaling mediates sperm preference in Drosophila females. in prep.
39. Avila FW, Cohen AB, Ameerudeen FS, Duneau D, Suresh S, Mattei AL, Wolfner MF. Retention of Ejaculate by Drosophila melanogaster Females Requires the Male-Derived Mating Plug Protein PEBme. Genetics. 2015;200(4):1171-9. doi: 10.1534/genetics.115.176669. PubMed PMID: 26058847.
40. Avila FW, Wong A, Sitnik JL, Wolfner MF. Don't pull the plug! the Drosophila mating plug preserves fertility. Fly. 2015;9(2):62-7. Epub 2015/12/24. doi: 10.1080/19336934.2015.1120931. PubMed PMID: 26695093; PMCID: PMC4826122.
41. Lupold S, Pitnick S, Berben KS, Blengini CS, Belote JM, Manier MK. Female mediation of competitive fertilization success in Drosophila melanogaster. Proceedings of the National Academy of Sciences of the United States of America. 2013;110(26):10693-8. Epub 2013/06/13. doi: 10.1073/pnas.1300954110. PubMed PMID: 23757499; PMCID: PMC3696778.
42. Lee KM, Daubnerova I, Isaac RE, Zhang C, Choi S, Chung J, Kim YJ. A neuronal pathway that controls sperm ejection and storage in female Drosophila. Current biology : CB. 2015;25(6):790-7. doi: 10.1016/j.cub.2015.01.050. PubMed PMID: 25702579.
43. Manier MK, Lupold S, Belote JM, Starmer WT, Berben KS, Ala-Honkola O, Collins WF, Pitnick S. Postcopulatory sexual selection generates speciation phenotypes in Drosophila. Current biology : CB. 2013;23(19):1853-62. Epub 2013/10/01. doi: 10.1016/j.cub.2013.07.086. PubMed PMID: 24076241.
44. Markow TA, Ankney PF. Insemination Reaction in Drosophila: Found in Species Whose Males Contribute Material to Oocytes before Fertilization. Evolution. 1988;42(5):1097-101. Epub 1988/09/01. doi: 10.1111/j.1558-5646.1988.tb02529.x. PubMed PMID: 28581170.
45. Laturney M, Billeter JC. Drosophila melanogaster females restore their attractiveness after mating by removing male anti-aphrodisiac pheromones. Nat Commun. 2016;7:12322. Epub 2016/08/04. doi: 10.1038/ncomms12322. PubMed PMID: 27484362; PMCID: PMC4976142.
46. Hopkins BR, Sepil I, Bonham S, Miller T, Charles PD, Fischer R, Kessler BM, Wilson C, Wigby S. BMP signaling inhibition in Drosophila secondary cells remodels the seminal proteome and self and rival ejaculate functions. Proceedings of the National Academy of Sciences of the United States of America.

2019;116(49):24719-28. Epub 2019/11/20. doi: 10.1073/pnas.1914491116. PubMed PMID: 31740617; PMCID: PMC6900634.
47. Lung O, Wolfner MF. Identification of and characterization of the major D. melanogaster mating plug protein. Ins Bioch Mol Bio. 2001;31:543-51.
48. Zeidler MP, Tan C, Bellaiche Y, Cherry S, Hader S, Gayko U, Perrimon N. Temperature-sensitive control of protein activity by conditionally splicing inteins. Nat Biotechnol. 2004;22(7):871-6. Epub 2004/06/09. doi: 10.1038/nbt979. PubMed PMID: 15184905.
49. Ogienko AA, Andreyeva EN, Omelina ES, Oshchepkova AL, Pindyurin AV. Molecular and cytological analysis of widely-used Gal4 driver lines for Drosophila neurobiology. BMC Genet. 2020;21(Suppl 1):96. Epub 2020/10/24. doi: 10.1186/s12863-020-00895-7. PubMed PMID: 33092520; PMCID: PMC7583314.
50. Cole SH, Carney GE, McClung CA, Willard SS, Taylor BJ, Hirsh J. Two functional but noncomplementing Drosophila tyrosine decarboxylase genes: distinct roles for neural tyramine and octopamine in female fertility. The Journal of biological chemistry. 2005;280(15):14948-55. PubMed PMID: 15691831.
51. Lee HG, Rohila S, Han KA. The octopamine receptor OAMB mediates ovulation via Ca2+/calmodulin-dependent protein kinase II in the Drosophila oviduct epithelium. PloS one. 2009;4(3):e4716. Epub 2009/03/06. doi: 10.1371/journal.pone.0004716. PubMed PMID: 19262750; PMCID: 2650798.
52. Poe AR, Wang B, Sapar ML, Ji H, Li K, Onabajo T, Fazliyeva R, Gibbs M, Qiu Y, Hu Y, Han C. Robust CRISPR/Cas9-Mediated Tissue-Specific Mutagenesis Reveals Gene Redundancy and Perdurance in Drosophila. Genetics. 2019;211(2):459-72. Epub 2018/12/07. doi: 10.1534/genetics.118.301736. PubMed PMID: 30504366; PMCID: PMC6366929.
53. Chow CY, Wolfner MF, Clark AG. The genetic basis for male x female interactions underlying variation in reproductive phenotypes of Drosophila. Genetics. 2010;186(4):1355-65. doi: 10.1534/genetics.110.123174. PubMed PMID: 20876561; PMCID: 2998317.
54. Monastirioti M, Linn CE, White K. Characterization of Drosophila tyramine beta-hydroxylase gene and isolation of mutant flies lacking octopamine. Journal of Neuroscience. 1996;16:3900-11.
55. Sweeney ST, Broadie K, Keane J, Niemann H, O'Kane CJ. Targeted expression of tetanus toxin light chain in Drosophila specifically eliminates synaptic transmission and causes behavioral defects. Neuron. 1995;14(2):341-51. Epub 1995/02/01. doi: 10.1016/0896-6273(95)90290-2. PubMed PMID: 7857643.
56. Kitamoto T. Conditional Modification of Behavior in Drosophila by Targeted Expression of a Temperature-Sensitive shibire Allele in Defined Neurons. J Neurobiol. 2001;47:81-92.
57. Hamada FN, Rosenzweig M, Kang K, Pulver SR, Ghezzi A, Jegla TJ, Garrity PA. An internal thermal sensor controlling temperature preference in Drosophila. Nature. 2008;454(7201):217-20. doi: 10.1038/nature07001. PubMed PMID: 18548007; PMCID: 2730888.
58. Bretman A, Lawniczak MK, Boone J, Chapman T. A mating plug protein reduces early female remating in Drosophila melanogaster. Journal of insect physiology. 2010;56(1):107-13. Epub 2009/10/06. doi: 10.1016/j.jinsphys.2009.09.010. PubMed PMID: 19800888.
59. Chapman T, Neubaum DM, Wolfner MF, Partridge L. The role of male accessory gland protein Acp36DE in sperm competition in Drosophila melanogaster. Proc Roy Soc Lond. 2000;267:1097-105.
60. Wong A, Albright SN, Giebel JD, Ram KR, Ji S, Fiumera AC, Wolfner MF. A role for Acp29AB, a predicted seminal fluid lectin, in female sperm storage in Drosophila melanogaster. Genetics. 2008;180(2):921-31. Epub 2008/09/02. doi: genetics.108.092106 [pii] 10.1534/genetics.108.092106. PubMed PMID: 18757944; PMCID: 2567391.
61. Mueller JL, Linklater JR, Ravi Ram K, Chapman T, Wolfner MF. Targeted gene deletion and phenotypic analysis of the Drosophila melanogaster seminal fluid protease inhibitor Acp62F. Genetics. 2008;178(3):1605-14. Epub 2008/02/05. doi: genetics.107.083766 [pii] 10.1534/genetics.107.083766. PubMed PMID: 18245332; PMCID: 2278106.
62. King EG, Macdonald SJ, Long AD. Properties and power of the Drosophila Synthetic Population Resource for the routine dissection of complex traits. Genetics. 2012;191(3):935-49. Epub 2012/04/17. doi: 10.1534/genetics.112.138537. PubMed PMID: 22505626; PMCID: PMC3389985.

63. Dietzl G, Chen D, Schnorrer F, Su KC, Barinova Y, Fellner M, Gasser B, Kinsey K, Oppel S, Scheiblauer S, Couto A, Marra V, Keleman K, Dickson BJ. A genome-wide transgenic RNAi library for conditional gene inactivation in Drosophila. Nature. 2007;448(7150):151-6. Epub 2007/07/13. doi: nature05954 [pii] 10.1038/nature05954. PubMed PMID: 17625558.
64. Ni JQ, Liu LP, Binari R, Hardy R, Shim HS, Cavallaro A, Booker M, Pfeiffer BD, Markstein M, Wang H, Villalta C, Laverty TR, Perkins LA, Perrimon N. A Drosophila resource of transgenic RNAi lines for neurogenetics. Genetics. 2009;182(4):1089-100. Epub 2009/06/03. doi: 10.1534/genetics.109.103630. PubMed PMID: 19487563; PMCID: 2728850.
65. Ravi Ram K, Wolfner MF. Sustained Post-Mating Response in Drosophila melanogaster Requires Multiple Seminal Fluid Proteins. PLoS genetics. 2007;3(12):e238. PubMed PMID: 18085830.
66. Singh A, Buehner NA, Lin H, Baranowski KJ, Findlay GD, Wolfner MF. Long-term interaction between Drosophila sperm and sex peptide is mediated by other seminal proteins that bind only transiently to sperm. Insect Biochem Mol Biol. 2018;102:43-51. Epub 2018/09/16. doi: 10.1016/j.ibmb.2018.09.004. PubMed PMID: 30217614; PMCID: PMC6249070.
67. Zirin J, Hu Y, Liu L, Yang-Zhou D, Colbeth R, Yan D, Ewen-Campen B, Tao R, Vogt E, VanNest S, Cavers C, Villalta C, Comjean A, Sun J, Wang X, Jia Y, Zhu R, Peng P, Yu J, Shen D, Qiu Y, Ayisi L, Ragoowansi H, Fenton E, Efrem S, Parks A, Saito K, Kondo S, Perkins L, Mohr SE, Ni J, Perrimon N. Large-Scale Transgenic Drosophila Resource Collections for Loss- and Gain-of-Function Studies. Genetics. 2020;214(4):755-67. Epub 2020/02/20. doi: 10.1534/genetics.119.302964. PubMed PMID: 32071193; PMCID: PMC7153935.
68. Avila FW, Wolfner MF. Acp36DE is required for uterine conformational changes in mated Drosophila females. Proceedings of the National Academy of Sciences of the United States of America. 2009;106(37):15796-800. Epub 2009/10/07. doi: 0904029106 [pii] 10.1073/pnas.0904029106. PubMed PMID: 19805225; PMCID: 2747198.
69. Mattei AL, Riccio ML, Avila FW, Wolfner MF. Integrated 3D view of postmating responses by the Drosophila melanogaster female reproductive tract, obtained by micro-computed tomography scanning. Proceedings of the National Academy of Sciences of the United States of America. 2015;112(27):8475-80. doi: 10.1073/pnas.1505797112. PubMed PMID: 26041806; PMCID: 4500220.
70. Schlamp F, Delbare SYN, Early AM, Wells MT, Basu S, Clark AG. Dense time-course gene expression profiling of the Drosophila melanogaster innate immune response. BMC genomics. 2021;22(1):304. Epub 2021/04/28. doi: 10.1186/s12864-021-07593-3. PubMed PMID: 33902461; PMCID: PMC8074482.
71. Heifetz Y, Lung O, Frongillo EA, Wolfner MF. The Drosophila seminal fluid protein Acp26Aa stimulates release of oocytes by the ovary. Curr Biol. 2000;10:99-102.
72. Peng J, Zipperlen P, Kubli E. Drosophila sex-peptide stimulates female innate immune system after mating via the Toll and Imd pathways. Current biology : CB. 2005;15(18):1690-4. PubMed PMID: 16169493.
73. Diaz F, Allan CW, Markow TA, Bono JM, Matzkin LM. Gene expression and alternative splicing dynamics are perturbed in female head transcriptomes following heterospecific copulation. BMC genomics. 2021;22(1):359. Epub 2021/05/20. doi: 10.1186/s12864-021-07669-0. PubMed PMID: 34006224; PMCID: PMC8132402.
74. Anreiter I, Sokolowski MB. The foraging Gene and Its Behavioral Effects: Pleiotropy and Plasticity. Annu Rev Genet. 2019;53:373-92. Epub 2019/09/06. doi: 10.1146/annurev-genet-112618-043536. PubMed PMID: 31487469.
75. Labeur C, Dallerac R, Wicker-Thomas C. Involvement of desat1 gene in the control of Drosophila melanogaster pheromone biosynthesis. Genetica. 2002;114(3):269-74. Epub 2002/09/11. doi: 10.1023/a:1016223000650. PubMed PMID: 12206365.
76. Bousquet F, Nojima T, Houot B, Chauvel I, Chaudy S, Dupas S, Yamamoto D, Ferveur JF. Expression of a desaturase gene, desat1, in neural and nonneural tissues separately affects perception and emission of sex pheromones in Drosophila. Proceedings of the National Academy of Sciences of the United States of America. 2012;109(1):249-54. Epub 2011/11/25. doi: 10.1073/pnas.1109166108. PubMed PMID: 22114190; PMCID: PMC3252891.

77. Patro R, Duggal G, Love MI, Irizarry RA, Kingsford C. Salmon provides fast and bias-aware quantification of transcript expression. Nature methods. 2017;14(4):417-9. Epub 2017/03/07. doi: 10.1038/nmeth.4197. PubMed PMID: 28263959; PMCID: PMC5600148.
78. Bhardwaj V, Semplicio G, Erdogdu NU, Manke T, Akhtar A. MAPCap allows high-resolution detection and differential expression analysis of transcription start sites. Nat Commun. 2019;10(1):3219. Epub 2019/08/01. doi: 10.1038/s41467-019-11115-x. PubMed PMID: 31363093; PMCID: PMC6667505.
79. Hu Q, Wolfner MF. The Drosophila Trpm channel mediates calcium influx during egg activation. Proceedings of the National Academy of Sciences of the United States of America. 2019;116(38):18994-9000. Epub 2019/08/21. doi: 10.1073/pnas.1906967116. PubMed PMID: 31427540; PMCID: PMC6754564.
80. Burtis KC, Baker BS. Drosophila doublesex gene controls somatic sexual differentiation by producing alternatively spliced mRNAs encoding related sex-specific polypeptides. Cell. 1989;56(6):997-1010. Epub 1989/03/24. doi: 10.1016/0092-8674(89)90633-8. PubMed PMID: 2493994.
81. Garaulet DL, Moro A, Lai EC. A double-negative gene regulatory circuit underlies the virgin behavioral state. Cell reports. 2021;36(1):109335. Epub 2021/07/08. doi: 10.1016/j.celrep.2021.109335. PubMed PMID: 34233178.
82. Anreiter I, Kramer JM, Sokolowski MB. Epigenetic mechanisms modulate differences in Drosophila foraging behavior. Proceedings of the National Academy of Sciences of the United States of America. 2017;114(47):12518-23. Epub 2017/10/29. doi: 10.1073/pnas.1710770114. PubMed PMID: 29078350; PMCID: PMC5703298.
83. Misra S, Wolfner M. The Drosophila seminal Sex Peptide can associate with rival as well as own 1 sperm and provide function for SP in polyandrous females. . BioRXiv 2020;doi: https://doi.org/10.1101/2020.02.20.958108.
84. Apger-McGlaughon J, Wolfner MF. Post-mating change in excretion by mated Drosophila melanogaster females is a long-term response that depends on sex peptide and sperm. Journal of insect physiology. 2013;59(10):1024-30. doi: 10.1016/j.jinsphys.2013.07.001. PubMed PMID: 23891750; PMCID: 3923306.
85. Cattelan M. Models for Paired Comparison Data: A Review with Emphasis on Dependent Data. Statist Sci 2012;27:412-33
86. Király FJ, Qian Z. Modelling competitive sports: Bradley-Terry-Élő models for supervised and on-line learning of paired competition outcomes. arXiv. 2017;1701.08055.
87. Élő A. The Rating of Chessplayers, Past and Present.: Arco; 1978.
88. Shafir S. Intransitivity of preferences in honey bees: support for “comparative” evaluation of foraging options. Animal Behaviour 48:55–67. Animal Behaviour. 1994;48:55–67.
89. McNamara JM, Trimmer PC, Houston AI. Natural selection can favour 'irrational' behaviour. Biol Lett. 2014;10(1):20130935. Epub 2014/01/17. doi: 10.1098/rsbl.2013.0935. PubMed PMID: 24429682; PMCID: PMC3917337.
90. Shev A, Fujii K, Hsieh F, McCowan B. Systemic testing on Bradley-Terry model against nonlinear ranking hierarchy. PloS one. 2014;9(12):e115367. Epub 2014/12/23. doi: 10.1371/journal.pone.0115367. PubMed PMID: 25531899; PMCID: PMC4274013.
91. Muller J, Wende B, Strob C, Eugster M, Gallenberger I, Floren A, Steffan-Dewenter I, Linsenmair KE, Weisser WW, Gossner MM. Forest management and regional tree composition drive the host preference of saproxylic beetle communities. . Jour Applied Ecol 2015;52:753-62.
92. Arbuthnott D, Fedina TY, Pletcher SD, Promislow DE. Mate choice in fruit flies is rational and adaptive. Nat Commun. 2017;8:13953. Epub 2017/01/18. doi: 10.1038/ncomms13953. PubMed PMID: 28094789; PMCID: PMC5247575.
93. Langville AN, Meyer CD. Who's #1?: The Science of Rating and Ranking Princeton NJ: Princeton University Press; 2012.
94. Gaertner BE, Ruedi EA, McCoy LJ, Moore JM, Wolfner MF, Mackay TF. Heritable variation in courtship patterns in Drosophila melanogaster. G3. 2015;5(4):531-9. Epub 2015/02/05. doi: 10.1534/g3.114.014811. PubMed PMID: 25650358; PMCID: PMC4390569.

95. Avila FW, Ravi Ram K, Bloch Qazi MC, Wolfner MF. Sex peptide is required for the efficient release of stored sperm in mated Drosophila females. Genetics. 2010;186(2):595-600. Epub 2010/08/04. doi: genetics.110.119735 [pii]10.1534/genetics.110.119735. PubMed PMID: 20679516; PMCID: 2954482.
96. Neubaum DM, Wolfner MF. Drosophila females require a seminal fluid protein, Acp36DE, to store sperm efficiently. Genetics. 1999;153:845-57.
97. Clark AG. Sperm competition and the maintenance of polymorphism. Heredity. 2002;88(2):148-53. Epub 2002/04/05. doi: 10.1038/sj.hdy.6800019. PubMed PMID: 11932773.



Ecology & Evolutionary Biology
UNIVERSITY OF TORONTO

Drs. Andrew Clark and Mariana Wolfner
Dept. of Molecular Biology and Genetics
Cornell University
Ithaca NY 14853

Dear Andy and Mariana,

I am writing to express my support for your grant proposal "Regulation of Gamete Use and Neural Pathways in Reproduction" and my willingness to provide mutants, reagents, and advice for the *Drosophila foraging*-gene-focused experiments you propose in Aim 2 of your grant.

My lab identified the *foraging (for)* gene years ago, and subsequently revealed and dissected its role in feeding, searching, learning, and other behaviors in *Drosophila*. We showed that *for* encodes a cGMP-dependent protein kinase that is expressed in the brain and elsewhere, and defined its promoter- and splice-isoforms and their roles in its expression patterns and functions. We also showed that *for* homologs carry out analogous functions in many other organisms, including humans. I have enjoyed hearing about your exciting finding of mating-regulation of *for*'s promoter- and splice-site use in the brain. The isoforms you find as mating-regulated make sense with our findings of the isoforms' functions in behaviors – consistent with the post-mating and preference behaviors that you are postulating for them.

I am eager to continue discussing your interesting results on *for* regulation and functions post-mating. My lab has generated many *for* mutant, transgene, and marker lines. I am happy to provide you with any of these strains, reagents, or advice that would be helpful for your proposed studies.

Best wishes,

MB Sokolowski

Marla B. Sokolowski, Ph.D., F.R.S.C.
University Professor,
Senior Fellow of Massey College,
Weston Fellow of CIFAR,
Departments of Ecology and Evolutionary Biology and Cell and Systems Biology,
University of Toronto,
25 Willcocks St,
Toronto, Ontario, Canada,
M5S 3B2
email: marla.sokolowski@utoronto.ca

RESOURCE SHARING PLAN

1. DATA-SHARING, FOR GRANTS WITH DIRECT COSTS > $500,000

Not applicable.

2. MODEL ORGANISMS

We are committed to the open sharing of Drosophila strains (mutants, transgenics and strains with wild-derived chromosomes), reagents (constructs, antibodies, databases, and sequences) and methods/protocols. During the past >30 years that each of our labs has been in existence, we have always made any strains, protocols or reagents that we generate freely available to anyone who requested them. These include fly lines with wild-derived chromosomes (AC) and transgenic lines expressing, knocking down, or mutant in reproductive or other proteins (MW), that are described in the proposal (other than fly lines that are already in the public domain). We will continue free and timely distribution of these items to all who request them, adhering to NIH Sharing Policies and Guidelines on NIH-Funded Research Resources (https://grants.nih.gov/policy/sharing.htm). We assure reviewers that we will follow the NIH guidelines that state that all transfers be made with no more restrictive terms than in the Simple Letter Agreement (SLA) or the Uniform Biological Materials Transfer Agreement (UBMTA) and without reach-through requirements, and that should any intellectual property arise that requires a patent, we will ensure that the technologies (materials and data) remain widely available to the research community in accordance with the NIH Policies and Guidelines.

3. GENOMIC DATA

Not applicable.

AUTHENTICATION OF KEY BIOLOGICAL AND/OR CHEMICAL RESOURCES

The key biological resources for this project are standard Drosophila strains, maintained in culture in our labs. Mutant and transgenic strains carry visible markers that allow rapid initial verification that they are of the correct identity; in some cases, the lines have visible phenotypes (e.g. green or red sperm-heads, mutant eye-colors (e.g. *w, v, cn,bw* and bw^{D})) that we use to confirm that the line is as expected, before each experiment. In addition we routinely carry out PCR or Western blotting (as appropriate) to verify that the strains contain the expected transgenes or are otherwise as expected. GAL4 drivers are checked periodically by crossing to UAS-GFP to confirm that they are driving expression where expected. DGRP and GDL lines are verified whenever we do RNA-seq, confirming that coding SNPs verify the line identities. Lines created by CRISPR are crossed into standard genetic backgrounds like w^{1118} and the presence of the expected mutation verified by sequencing of genomic DNA and/or by PCR assays. Knockdowns are verified by RT-PCR, supplemented with or replaced by Western blotting to examine the level of the targeted protein, where appropriate. All antibodies are checked for specificity on Western blots (or by immunofluorescence where appropriate) prior to experimental use, verifying that they recognize a protein of correct size in wildtype and that absent or reduced in amount in mutant or knockdown animals. All DNA constructs are validated by sequencing prior to use or introduction into flies or E. coli. RNA-seq methods and analysis, and the statistical methods used to assess sperm competition and other results are all standard ones that our labs have previously used in published peer-reviewed studies. Our experiments use only standard laboratory-chemical reagents (buffers, etc.). We do however always test them (e.g. test pH of buffers, etc.) prior to use.